

NATIONAL ECONOMICS UNIVERSITY
FACULTY OF MATHEMATICAL ECONOMICS



GRADUATION THESIS

Enhancing Course Recommendation via Domain-Adaptive Contrastive Learning and Graph-based Reasoning

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Hanoi - May 2026

ACKNOWLEDGEMENT

This thesis is the result of a long and challenging journey, one that would not have been possible without the invaluable support and guidance of numerous individuals.

First and foremost, I would like to express my deepest gratitude to my supervisor, Dr. Tran Hung from the Faculty of Mathematical Economics, National Economics University. I am truly fortunate to have had his guidance. Beyond his professional expertise, he also guided me in maintaining academic integrity and research rigor.

I am also grateful to all the lecturers at the Faculty of Mathematical Economics. The knowledge and inspiration I received during my years at NEU have been the foundation for this work and my future career.

To my family and friends, thank you for being my constant source of motivation. Your belief in me, even at times when I doubted myself, was the greatest gift I could ask for.

Finally, while I have put my best effort into this thesis, I am aware that it may still have shortcomings. I look forward to receiving feedback from my teachers and peers so that I can continue to learn and grow.

DECLARATION

I certify that this thesis is my own original work, conducted independently and without any prohibited assistance. All external sources, data, and literature used in this research have been fully acknowledged and correctly cited in accordance with academic integrity regulations. Furthermore, I confirm that this work has not been previously submitted, either in whole or in part, for any other degree or qualification at any university.

Hanoi, 05/2026

Signature

Nguyen Duy Hung

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LIST OF ABBREVIATIONS

Abbreviation	Full Meaning
AI	Artificial Intelligence
AP	Average Precision
BERT	Bidirectional Encoder Representations from Transformers
CBF	Content-based Filtering
CBOW	Continuous Bag-of-Words
CF	Collaborative Filtering
CI	Confidence Interval
CLO	Course Learning Outcomes
CUDA	Compute Unified Device Architecture
CV	Curriculum Vitae
DAG	Directed Acyclic Graph
DAPT	Domain-Adaptive Pre-Training
DFS	Depth-First Search
DPR	Dense Passage Retrieval
E2E	End-to-End
ELMo	Embeddings from Language Models
ESCO	European Skills, Competences, Qualifications and Occupations
GGUF	GPT-Generated Unified Format
GNN	Graph Neural Network
GPT	Generative Pre-trained Transformer
GPU	Graphics Processing Unit

Abbreviation	Full Meaning
IDCG	Ideal Discounted Cumulative Gain
JD	Job Description
KG	Knowledge Graph
KGE	Knowledge Graph Embedding
LLM	Large Language Model
LSTM	Long Short-Term Memory
MAP	Mean Average Precision
MMTEB	Massive Multilingual Text Embedding Benchmark
MOOC	Massive Open Online Courses
MRR	Mean Reciprocal Rank
NDCG	Normalized Discounted Cumulative Gain
NLP	Natural Language Processing
PCGNN	Prerequisite-aware Category-aware Graph Neural Network
PEFT	Parameter-Efficient Fine-Tuning
PLM	Pre-trained Language Model
QA	Question Answering
RLHF	Reinforcement Learning from Human Feedback
RNN	Recurrent Neural Network
RoBERTa	Robustly Optimized BERT Approach
RR	Reciprocal Rank
URI	Unique Identifier
VRAM	Video Random Access Memory

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ABSTRACT

Skill-gap-based course recommendation is an important problem for supporting learners and job seekers in selecting courses that align with their target career goals. This thesis proposes an end-to-end course recommendation framework that maps job requirements and candidate profiles into missing skills, and then recommends suitable courses to address those gaps. The framework combines domain-adaptive contrastive learning with graph-based reasoning. First, a domain-specific synthetic triplet dataset is constructed from skill taxonomies and LLM-generated skill variations, with hard negatives used to improve fine-grained skill discrimination. A Qwen3-based embedding model is then fine-tuned using contrastive learning to improve semantic matching between skill expressions and course-related content. Second, a knowledge graph is built to represent courses, skills, and pedagogical relations through TEACHES and REQUIRES edges, enabling prerequisite-aware retrieval and course sequencing. The system is evaluated using both skill-matching and end-to-end recommendation metrics, including Hit Rate, Recall, NDCG, MRR, and MAP. Experimental results show that the fine-tuned embedding model improves ranking quality compared with the pre-trained backbone, while the KG-augmented recommendation pipeline achieves stronger end-to-end performance than basic embedding search. In the final recommendation setting, the proposed system achieves Hit Rate@1 of 0.590 and Hit Rate@5 of 0.920. These results suggest that combining domain-adapted semantic representations with structured graph reasoning is a promising direction for career-oriented course recommendation. However, the current evaluation is limited by the size of the end-to-end test set and the reliance on synthetic data, motivating future work on larger-scale validation, user studies, and ablation analysis of graph constraints.

CHAPTER 1: INTRODUCTION

1.1 The significance of the topic

Recent reports from many international organizations show that online education has expanded to an unprecedented scale. By 2021, massive open online courses (MOOCs) had reached over 220 million learners globally (Shah, 2021). At the same time, leading platforms have attracted tens of millions of users. For example, Coursera reported approximately 142 million registered learners by the end of 2023 (Coursera, Inc., 2024). These numbers clearly illustrate the massive scale of online education, confirming that digital learning has become a mainstream educational pathway rather than a supplementary one. This rapid growth is fundamentally driven by the accelerating pace of skill obsolescence in the modern labor market. The World Economic Forum estimated that 44% of workers' core skills will be disrupted by 2027 due to the rapid integration of emerging technologies (World Economic Forum, 2023). This growth is further propelled by the relentless emergence of novel professional competencies. According to the LinkedIn's Economic Graph (LinkedIn Learning, 2025), the skill sets required for global occupations have evolved significantly, with a projected change of 68% by 2030—an acceleration driven largely by the integration of generative AI into the workforce. As a result, the digital learning landscape is constantly changing with new topics, requiring learners to master skills that did not even exist in traditional academic programs just a few years ago.

However, the massive expansion of course catalogs has led to serious information overload. Research has shown that learners in MOOC environments struggle to identify suitable courses as the number of available options increases dramatically. One direct consequence is low engagement. Studies report that only 7–13% of enrolled learners actually complete MOOC courses (Nesterowicz et al., 2022), indicating that many users either choose unsuitable courses or lose interest after enrollment. Traditional recommendation approaches — mainly collaborative filtering or simple content matching — rely heavily on past interactions and therefore struggle when user preferences are unclear or when courses are newly introduced. Consequently, recommendation quality remains limited and learner motivation decreases.

To overcome these challenges, this research leverages the potential of contrastive learning and graph-based reasoning. Specifically, contrastive learning is employed to refine representation quality by filtering inherent noise and capturing semantic similarities between courses, even those spanning diverse subject areas. This allows models to achieve better performance in cold-start scenarios where interaction data is minimal. On the other hand, graph-based reasoning enhances the system by modeling semantic connections

between learners, courses and skills rather than depending only on past interactions. This approach enables the system to infer relationships and support recommendations that actually consider the prerequisite relationship between courses. Consequently, these advanced methods not only improve recommendation accuracy but also ensure explainability, as recommended items can be traced through meaningful relational paths.

1.2 Overview of previous studies

Recommender systems have been widely studied to address the problem of information overload in online platforms. Over the past two decades, the field has evolved through several major paradigms, ranging from rule-based approaches to modern deep learning and graph-based recommendation models.

Early recommender systems were primarily based on rule-based or knowledge-based approaches. These systems relied on manually designed rules and domain knowledge to match user profiles with suitable items. Although such systems were interpretable and easy to implement, they lacked adaptability and scalability when dealing with large and dynamic datasets.

Content-based filtering (CBF) was one of the earliest data-driven approaches (Lops et al., 2011). In this paradigm, items and user profiles are represented as vectors in a feature space. Formally, an item j is represented by a feature vector $\mathbf{x}_j$. A common weighting scheme is TF-IDF (Term Frequency-Inverse Document Frequency), which quantifies the relevance of a term t in a document d :

$$w_{t,d} = tf_{t,d} \times \log \frac{N}{df_t} \quad (1.1)$$

The system constructs a user profile $\mathbf{u}_i$ by aggregating vectors of interacted items. The recommendation score for an unseen item j is calculated using Cosine Similarity:

$$\text{score}(i, j) = \frac{\mathbf{u}_i \cdot \mathbf{x}_j}{\|\mathbf{u}_i\| \|\mathbf{x}_j\|} \quad (1.2)$$

Despite its effectiveness, CBF often suffers from overspecialization, recommending items too similar to past consumption (Lops et al., 2011).

Collaborative filtering (CF) later became the dominant paradigm (Koren et al., 2009). CF leverages the collective behavior of similar users to predict preferences. It is centered around the User-Item Interaction Matrix $R_{m \times n}$. While neighborhood-based CF computes similarities between users or items, Matrix Factorization (MF) decomposes R into lower-dimensional latent matrices: User matrix $P \in \mathbb{R}^{m \times k}$ and Item matrix $Q \in \mathbb{R}^{n \times k}$:

$$\hat{r}_{u,i} = \mathbf{p}_u \cdot \mathbf{q}_i^T \quad (1.3)$$

The latent factors are learned by minimizing a regularized squared error loss:

$$\mathcal{L} = \sum_{(u,i) \in \mathcal{K}} (r_{u,i} - \mathbf{p}_u \mathbf{q}_i^T)^2 + \lambda (\|\mathbf{p}_u\|^2 + \|\mathbf{q}_i\|^2) \quad (1.4)$$

Traditional CF methods, however, struggle with data sparsity and cold-start problems (Koren et al., 2009). To overcome these, hybrid recommender systems combine CF with content-based signals to improve accuracy and coverage (Burke, 2002).

Despite their success, traditional CF methods suffer from several limitations, including data sparsity and cold-start problems. To overcome these issues, hybrid recommender systems were proposed to combine collaborative filtering with content-based signals (Burke, 2002). Hybrid models integrate multiple information sources to improve recommendation accuracy and coverage.

With the rapid growth of deep learning, neural recommender systems have been proposed to model complex user-item interactions. These approaches utilize neural networks to learn high-dimensional representations of users and items from implicit feedback and contextual information. Recent research has shown that deep learning models can capture nonlinear relationships in recommendation data and outperform traditional models in many scenarios (Gheewala et al., 2025).

More recently, graph-based recommendation methods have emerged as a powerful paradigm. These methods model recommender systems as graphs consisting of users, items, and additional entities such as skills, topics, or knowledge concepts. Knowledge graph-based recommender systems incorporate structured relationships between entities to enhance recommendation quality (Wang et al., 2019a). By propagating information through graph structures, these models can capture higher-order relationships between users and items.

In addition, knowledge graph reasoning methods have been proposed to improve interpretability and performance. For example, path-based reasoning over knowledge graphs can explain the rationale behind recommendations and model semantic relationships between entities (Wang et al., 2019b). These approaches demonstrate the growing importance of structured knowledge in modern recommender systems.

Overall, the evolution of recommender systems reflects a shift from rule-based heuristics toward data-driven and knowledge-aware models. While recent graph-based approaches have shown promising results, several fundamental challenges remain unresolved.

1.3 Research gaps

Despite significant progress in recommender system research, several critical limitations remain when applying these technologies to the educational and career development domains. This research identifies three primary gaps that motivate the proposed framework:

First, **there is a severe scarcity of high-quality, labeled datasets for representation learning at the intersection of occupational taxonomies and academic curricula.** While recent efforts like the Course-Skill Atlas (Sabet et al., 2024) have successfully aligned millions of course syllabi with occupational frameworks like O*NET at scale, these datasets are primarily descriptive and lack the triplet structure (anchor, positive, and hard negative) required for contrastive learning. Furthermore, as noted by Frej et al. (Frej et al., 2024a), the community faces a shortage of public course-recommendation benchmarks, with only a few datasets like XuetaangX and COCO being widely accessible. Most existing data focus on simple user-item interactions rather than the nuanced semantic mapping between granular skills and educational content. This lack of specialized triplet data hinders the ability to train models that can distinguish between conceptually similar but functionally distinct professional competencies. This gap necessitates a framework capable of augmenting existing datasets with high-quality synthetic supervision signals, utilizing Large Language Models to bridge the data scarcity in educational contrastive learning.

Second, **general-purpose embedding models exhibit significant semantic limitations in specialized technical domains.** Foundation models such as BERT or RoBERTa are trained on broad corpora and often fail to capture fine-grained distinctions in technical terminology—for instance, confusing "Data Analysis" with "Data Management" or failing to distinguish between specific software tools within the same ecosystem. Research by Turchin et al. (Turchin et al., 2023) demonstrates that domain-specific variants (e.g., BioBERT) significantly outperform general models in specialized tasks, yet domain-adaptive fine-tuning remains under-explored in the context of course-to-skill matching. Furthermore, as highlighted by the work on AdaSent (Huang et al., 2023), capturing in-domain semantics typically requires a complex two-step process of Domain-Adaptive Pre-Training (DAPT) and supervised fine-tuning, which has not been effectively integrated into existing course recommendation pipelines. Therefore, this research proposes a domain-adaptive contrastive learning approach to bridge the semantic gap between general-purpose foundation models and the specialized requirements of technical professional competencies.

Third, **current graph-based recommendation systems lack explicit pedagogical reasoning and strict prerequisite constraints.** While some models, such as PCGNN

(Sun et al., 2024), have begun incorporating course prerequisites into Graph Neural Networks (GNNs), they often treat these relationships as latent signals to enhance embeddings rather than tools for explicit logical reasoning. Consequently, the "pedagogical logic" remains implicit. Furthermore, even state-of-the-art models (Frej et al., 2024b) typically prioritize relational similarity (e.g., courses in the same category) over logical progression. As argued by Hu et al. (Hu et al., 2022), prerequisites are a fundamental constraint in education; without a mechanism to enforce that foundational skills are acquired before advanced modules, recommended learning paths may be semantically relevant but logically incoherent and practically unfeasible for the learner. To address this, we introduce a Knowledge Graph-based reasoning layer that explicitly encodes pedagogical constraints, ensuring that recommended course sequences move beyond simple similarity to achieve logical and practical feasibility.

1.4 Objectives and Research Questions

1.4.1 Objectives

The main objective of this research is to build a complete course recommendation system that helps individuals advance in their professional career paths. To achieve this, the study focuses on the following specific goals:

- **To create a high-quality, domain-specific triplet dataset** for skill-to-course semantic alignment. By utilizing Large Language Models (LLMs) to generate synthetic phrasings and challenging hard negatives, this research aims to mitigate the current scarcity of labeled data required for contrastive representation learning.
- **To develop a domain-adaptive embedding model** for specialized skills and qualifications. This model will be fine-tuned on domain-specific documents to capture nuanced skill semantics that general-purpose language models often miss.
- **To construct a Knowledge Graph (KG) reasoning framework** that formalizes educational hierarchies. This goal focuses on modeling entities (Courses, Skills) and explicitly encoding pedagogical relationships (TEACHES and REQUIRES) to ensure logical progression in recommendations.
- **To implement and validate a comprehensive evaluation pipeline** using standard ranking metrics, including Hit Rate, NDCG, MRR, and MAP. This objective ensures an objective measurement of recommendation accuracy and benchmarks the framework's performance.

1.4.2 Research Questions

To evaluate the effectiveness of the proposed framework, this research addresses the following questions:

1. How can the generation of synthetic triplets and hard negatives through LLMs effectively provide the supervision signals necessary to train robust skill-based embedding models in data-scarce environments?
2. How can domain-adaptive contrastive learning improve the model's ability to capture fine-grained semantic distinctions between specialized professional competencies compared to general-purpose foundation models?
3. How does the integration of explicit prerequisite constraints within a knowledge-graph-based reasoning framework enhance the logical coherence, ordering, and practical feasibility of recommended course sequences for a learner?

CHAPTER 2: LITERATURE REVIEW

2.1 Large Language Models

2.1.1 *Definition and Conceptual Framework*

Large Language Models (LLMs) represent an advanced class of artificial intelligence systems designed to process, interpret, and generate human-like text by learning from massive datasets. Unlike traditional Natural Language Processing (NLP) models that were often task-specific, LLMs are characterized by their immense scale—often containing billions of parameters—and their training on diverse, large-scale corpora (Naveed et al., 2025). At their technical core, these models are deep neural networks that function as probabilistic engines, calculating the likelihood of a subsequent token in a sequence based on the preceding context. A defining trait of modern LLMs is the emergence of "emergent abilities," such as complex reasoning and in-context learning, which typically manifest only when the model reaches a certain threshold of computational scale and data volume (Naveed et al., 2025).

2.1.2 *Historical Evolution and the Transformer Breakthrough*

The development of Natural Language Processing (NLP) has transitioned from early statistical methods to deep neural architectures designed to capture the complexities of human language. For years, sequence modeling was dominated by Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which processed text tokens sequentially. However, these architectures were fundamentally limited by their inability to parallelize training and their struggle to maintain long-range dependencies in lengthy documents (Vaswani et al., 2017). A paradigm shift occurred with the introduction of the Transformer architecture, which replaced recurrence with the self-attention mechanism, enabling massive parallelization and more efficient representation learning. This innovation laid the groundwork for Pre-trained Language Models (PLMs) like BERT and early GPT variants, which utilized self-supervised learning on massive corpora. As these models were scaled to billions of parameters, they evolved into Large Language Models (LLMs), demonstrating emergent abilities such as zero-shot reasoning and complex instruction following that were not present in smaller-scale predecessors (Brown et al., 2020; Naveed et al., 2025).

2.1.3 *Mathematical Foundations of the Attention Mechanism*

The operational efficiency of LLMs is primarily derived from the *Scaled Dot-Product Attention* mechanism. This mathematical framework allows the model to assign

varying degrees of importance to different parts of the input sequence regardless of their distance (Vaswani et al., 2017).

The process begins by mapping each input token into three distinct vectors: a Query (Q), a Key (K), and a Value (V). For a set of queries packed into a matrix, the attention output is calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2.1)$$

In this formulation, d_k represents the dimensionality of the keys. The dot product QK^T computes the raw alignment or compatibility scores between tokens. A critical technical detail is the scaling factor $1/\sqrt{d_k}$. As d_k increases, the magnitude of the dot products grows, which can push the softmax function into regions where its output is extremely close to 0 or 1. In these regions, the gradients of the function become vanishingly small (a phenomenon known as *softmax saturation*), which severely hinders the model’s ability to learn during backpropagation. Scaling the dot products ensures that the input to the softmax remains within a range that maintains numerical stability and healthy gradient flow (Vaswani et al., 2017).

To enhance the model’s ability to focus on different positions and representation subspaces simultaneously, LLMs employ *Multi-Head Attention*. This involves performing the attention function in parallel across h different projections:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (2.2)$$

where each head_i is an independent attention operation. This architectural design allows the model to integrate information from multiple contextual perspectives, which is crucial for understanding complex linguistic structures (Vaswani et al., 2017).

2.1.4 *Adaptation Regimes and Alignment*

The development of a functional LLM typically involves multiple stages. The primary stage is Pre-training, where the model learns fundamental language patterns through self-supervised objectives like Causal Language Modeling (predicting the next token) or Masked Language Modeling (Naveed et al., 2025).

However, pre-trained models may not always follow specific user instructions or may generate harmful content. To rectify this, models undergo Fine-tuning and Alignment. Techniques such as Instruction Tuning use human-curated prompt-response pairs to teach the model how to respond to queries. Furthermore, Reinforcement Learning from Human Feedback (RLHF) is employed to align model outputs with human values (helpfulness,

honesty, and harmlessness) by training a reward model to rank generated responses (Naveed et al., 2025).

2.1.5 Diverse Applications of LLMs

The versatility of LLMs has enabled their integration into numerous professional and creative domains:

- **General Purpose and Knowledge Management:** LLMs serve as advanced assistants for summarization, translation, and information retrieval, filtering vast amounts of data to provide concise insights (Naveed et al., 2025).
- **Software Development:** Specialized models like AlphaCode and Codex facilitate automated code generation, debugging, and multi-turn program synthesis, significantly augmenting developer productivity.
- **Professional Domains (Medicine, Law, Finance):** In healthcare, LLMs assist in clinical decision support and medical research synthesis. In the legal sector, they are used for thematic analysis of legal documents and case law explanation. Financial models like BloombergGPT utilize industry-specific datasets for sentiment analysis and algorithmic trading support (Naveed et al., 2025).

By combining the structural efficiency of the Transformer with massive scaling and sophisticated alignment techniques, LLMs have evolved from simple text predictors into powerful reasoning engines capable of reshaping modern technology.

2.1.6 LLMs for Synthetic Data Generation and Augmentation

A significant challenge in training specialized recommender systems is the scarcity of high-quality, labeled datasets, particularly for contrastive learning tasks. LLMs have emerged as a promising solution for synthetic data generation, enabling the creation of large-scale supervised signals without the prohibitive cost of human annotation. Synthetic data can support various machine learning paradigms, including data augmentation, self-supervised learning, and regularization, thereby improving model robustness and generalization (Fonseca and Bacao, 2023).

In the context of representation learning, LLMs are increasingly used to generate diverse "positive" samples and informative "hard negatives." Unlike simple augmentation techniques (e.g., synonym replacement), LLMs enable semantic paraphrasing, producing linguistically diverse yet semantically consistent samples that enrich the training distribution. This increased diversity can lead to more discriminative embeddings and improved downstream performance. In particular, synthetic hard negatives have been shown to

enhance contrastive learning by providing more challenging and informative training signals, thereby improving the model’s ability to distinguish subtle semantic differences (Dong et al., 2024).

Despite these advantages, LLM-based synthetic data generation also presents several limitations. First, the quality of synthetic data is inherently dependent on the generating model, which may introduce biases, factual inaccuracies, or distributional artifacts that deviate from real-world data. Second, synthetic data may fail to capture the full complexity and long-tail characteristics of real user behavior, potentially leading to reduced generalization in real-world deployment. Third, excessive reliance on synthetic data can amplify existing model biases or result in distributional drift, particularly when synthetic samples are recursively generated. Additionally, the computational cost of generating high-quality synthetic data using large models can be non-trivial, posing practical challenges for large-scale systems (Fonseca and Bacao, 2023).

2.2 Embedding Models

2.2.1 Definitions and History

Embedding models aim to represent discrete data (e.g., words, sentences, or entities) as continuous vectors in a dense, low-dimensional space, such that semantic similarity is reflected by geometric proximity.

Before the dominance of neural embeddings, text was primarily represented using sparse (or localist) methods such as one-hot encoding or Bag-of-Words (BoW). These approaches suffered from the curse of dimensionality, where the vector size increased linearly with the vocabulary size, and an inherent inability to capture semantic relationships between words, as all word vectors were orthogonal to each other. To overcome these limitations, the paradigm shifted toward distributed representations. The fundamental idea, rooted in the distributional hypothesis, is that words occurring in similar contexts tend to have similar meanings. Early efforts to learn such dense, continuous vector spaces include neural probabilistic language models (Bengio et al., 2003). Unlike sparse methods, these distributed models represent words in a much lower-dimensional manifold where semantic proximity is preserved through learned weights.

A major breakthrough came with the introduction of Static Word Embeddings (2013–2016) such as Word2Vec (Mikolov et al., 2013), which proposed efficient training objectives such as Skip-gram and Continuous Bag-of-Words (CBOW) to learn word embeddings from large corpora. These models demonstrated that semantic relationships could be captured through vector arithmetic. Shortly after, GloVe (Pennington et al., 2014) improved upon this by incorporating global word co-occurrence statistics, providing more stable and interpretable embeddings. Despite their effectiveness, these embeddings

are static, meaning each word has a single representation regardless of context.

The next major advancement addressed the limitations of static embeddings by introducing Contextualized Embeddings Models (2018–2019). ELMo (Peters et al., 2018) leveraged deep bidirectional language models to generate embeddings conditioned on the entire sentence. This idea was further advanced by BERT (Devlin et al., 2019), which employed the Transformer architecture (Vaswani et al., 2017) and self-supervised objectives such as masked language modeling. These models produce dynamic embeddings that vary depending on the surrounding context, significantly improving performance across a wide range of NLP tasks.

While contextual models provide powerful token-level representations, they are not inherently optimized for sentence-level similarity tasks. Sentence-BERT (Reimers and Gurevych, 2019) addressed this limitation by introducing a siamese network structure with contrastive learning objectives, enabling efficient computation of semantically meaningful sentence embeddings. Concurrently, dense retrieval models such as DPR (Karpukhin et al., 2020) adopted dual-encoder architectures to learn embeddings for queries and documents, facilitating scalable semantic search through vector similarity.

Recent advances have emphasized contrastive learning (2020–Present) as a central paradigm for embedding training. Frameworks such as SimCLR (Chen et al., 2020) demonstrated the effectiveness of learning representations by maximizing agreement between positive pairs while minimizing similarity to negative samples. This approach has been widely adopted in both vision and language domains, including models such as CLIP (Radford et al., 2021), which aligns text and image embeddings in a shared space.

In the context of language, modern embedding models increasingly rely on large-scale pretraining combined with contrastive fine-tuning to produce high-quality representations for tasks such as semantic matching, clustering, and deduplication. Recent works have also explored instruction-tuned and domain-specific embeddings, enabling improved performance in specialized applications such as skill representation and job matching.

2.2.2 Model Architectures

Modern embedding models are predominantly based on the Transformer architecture (Vaswani et al., 2017), which enables efficient modeling of contextual dependencies through self-attention mechanisms. Transformer-based encoders, such as BERT, generate contextualized token representations that can be aggregated (e.g., via pooling) to produce sentence-level embeddings.

Sentence embedding models typically adopt encoder-only architectures for effi-

ciency in retrieval scenarios. Variants such as Sentence-BERT (Reimers and Gurevych, 2019) employ siamese networks, where two or more input sequences are encoded independently using shared parameters, and their embeddings are compared using similarity metrics such as cosine similarity.

Recent advances have introduced specialized architectures and training strategies for embedding learning, including dual-encoder frameworks for large-scale retrieval systems, where query and document embeddings are computed independently to enable efficient nearest neighbor search (Karpukhin et al., 2020).

2.2.3 Training Regimes

The training of embedding models has evolved from unsupervised learning to more sophisticated regimes incorporating supervised and self-supervised signals. Early models such as Word2Vec and GloVe rely on unsupervised learning from raw text corpora (Mikolov et al., 2013; Pennington et al., 2014).

Contextual models like BERT are trained using self-supervised objectives on large-scale corpora, enabling them to learn rich linguistic representations without manual annotation (Devlin et al., 2019). However, for tasks requiring fine-grained semantic alignment, additional fine-tuning is often necessary.

Recent approaches emphasize contrastive learning frameworks, where models are trained on pairs or triplets of data to explicitly encode similarity relationships. This paradigm has been shown to significantly improve performance in semantic matching, clustering, and retrieval tasks (Chen et al., 2020; Karpukhin et al., 2020). Furthermore, domain-specific embedding models can be obtained by fine-tuning on task-specific datasets, allowing the embedding space to better capture domain-relevant semantics, such as skills or job-related concepts.

Overall, the combination of large-scale pretraining and task-specific contrastive fine-tuning has emerged as a dominant paradigm for learning high-quality embeddings suitable for downstream applications.

2.2.4 Applications

Semantic matching is one of the most prominent applications of embedding models, where the goal is to measure the similarity between two pieces of text based on meaning rather than surface form. Traditional lexical matching approaches, such as keyword-based retrieval, often fail to capture paraphrases or semantic equivalence. Embedding models address this limitation by representing semantically similar texts as nearby vectors in the embedding space.

Sentence-level embedding models, such as Sentence-BERT (Reimers and Gurevych, 2019), have been specifically designed to support efficient semantic similarity computation using distance metrics such as cosine similarity. These models are widely used in applications such as semantic search, question answering, and recommendation systems. In large-scale retrieval settings, dual-encoder architectures (Karpukhin et al., 2020) enable fast nearest-neighbor search, making them suitable for real-time matching tasks involving large corpora.

Embedding models also play a crucial role in clustering tasks, where the objective is to group semantically similar items without explicit supervision. By transforming textual data into dense vector representations, embeddings make it possible to apply standard clustering algorithms such as k-means or hierarchical clustering in a meaningful semantic space.

In this context, the quality of embeddings directly impacts the coherence and separability of clusters. Contextual and contrastively trained embeddings have been shown to significantly improve clustering performance by better capturing semantic relationships (Chen et al., 2020). Applications of clustering with embeddings include topic discovery, taxonomy construction, and organization of large-scale text collections, such as grouping similar skills or job descriptions into coherent categories.

Deduplication is another critical application of embedding models, particularly in domains with noisy or redundant data. The goal is to identify and remove duplicate or near-duplicate entries that may differ in wording but share the same semantic meaning.

Embedding-based deduplication relies on computing similarity scores between items and identifying pairs that exceed a predefined threshold. Unlike exact string matching, this approach is robust to variations in phrasing, spelling, and formatting. Recent embedding models trained with contrastive objectives are especially effective in this setting, as they explicitly enforce similarity between semantically equivalent inputs while separating unrelated ones.

This capability is particularly valuable in structured domains such as skill normalization, where the same skill may be expressed in multiple ways. Embedding-based deduplication enables the consolidation of such variations into canonical representations, improving data quality and downstream retrieval performance.

Overall, embedding models provide a unified framework for semantic matching, clustering, and deduplication by encoding meaning into a continuous vector space. Their effectiveness in these tasks has made them a cornerstone of modern retrieval and representation systems, particularly when combined with contrastive learning techniques that enhance the alignment of the embedding space with semantic similarity objectives.

2.3 Knowledge Graph

2.3.1 Definition

A Knowledge Graph (KG) is a structured representation of real-world entities and their relationships, typically modeled as a graph where nodes represent entities and edges represent relations between them. Formally, a KG is often defined as a set of triples (h, r, t) , where h (head entity) and t (tail entity) are connected by a relation r .

Knowledge Graphs are designed to integrate heterogeneous information into a unified representation, enabling machines to perform reasoning, semantic search, and knowledge discovery. Unlike traditional relational databases, KGs emphasize semantic relationships and flexible schema design, making them particularly suitable for complex, interconnected data (Hogan et al., 2021).

In recent years, Knowledge Graph Embedding (KGE) methods have been introduced to map entities and relations into continuous vector spaces while preserving the structural information of the graph. These embeddings facilitate downstream tasks such as link prediction, entity classification, and recommendation (Wang et al., 2017). Popular approaches include translational distance models such as TransE, which interpret relations as translations in embedding space (Bordes et al., 2013), as well as more advanced models like RotatE (Sun et al., 2019).

2.3.2 History

The concept of representing knowledge as graphs has its roots in early semantic networks and symbolic AI research in the 1970s and 1980s. However, modern Knowledge Graphs gained prominence with the rise of large-scale web data and the need for structured knowledge representation.

A major milestone was the introduction of large, publicly available Knowledge Graphs such as DBpedia (Auer et al., 2007), which extracted structured information from Wikipedia, and Freebase (Bollacker et al., 2008), a collaboratively curated knowledge base. These projects demonstrated the feasibility of building large-scale KGs and enabled widespread research in the field.

The term “Knowledge Graph” became widely popular after Google introduced its Knowledge Graph in 2012 to enhance search engine capabilities by providing semantic search results and entity-based answers. This marked a shift from keyword-based retrieval to entity-centric information systems.

Subsequently, research focus shifted toward representation learning on Knowledge Graphs. The introduction of TransE (Bordes et al., 2013) marked a turning point by enabling scalable embedding of large graphs. This was followed by a surge of embedding

models, including TransH, TransR, and RotatE (Sun et al., 2019), which improved the ability to model complex relations.

More recently, Knowledge Graph research has expanded into integration with deep learning and large language models, as well as applications in recommendation systems, question answering, and reasoning. Comprehensive surveys such as (Nickel et al., 2015; Wang et al., 2017; Hogan et al., 2021) summarize the evolution of the field and highlight ongoing challenges such as scalability, incompleteness, and dynamic updates.

2.3.3 Applications

Knowledge Graphs (KGs) have been widely adopted across a variety of domains due to their ability to represent structured, semantic relationships between entities. Their applications span multiple tasks in artificial intelligence, data mining, and information retrieval.

One of the most prominent applications of KGs is in search and information retrieval. By leveraging entity-centric representations, KGs enable semantic search that goes beyond keyword matching, allowing systems to understand user intent and return more relevant results. This paradigm has been successfully deployed in industrial systems such as web search engines (Hogan et al., 2021).

KGs also play a critical role in question answering (QA) systems. By structuring knowledge into triples, KGs allow QA systems to perform multi-hop reasoning over entities and relations to derive answers. Approaches range from symbolic traversal-based reasoning to embedding-based methods that rank candidate answers (Nickel et al., 2015).

Another important application is recommendation systems. Knowledge Graph-based recommendation methods enhance traditional collaborative filtering by incorporating rich semantic information about users, items, and their relationships. Techniques include path-based reasoning, meta-path exploration, and embedding-based approaches that learn latent representations of entities (Wang et al., 2019a). These methods improve recommendation accuracy, interpretability, and the ability to handle data sparsity.

In natural language processing (NLP), KGs are used to enhance language understanding by providing structured background knowledge. They are commonly integrated into tasks such as named entity recognition, entity linking, and text generation, where external knowledge improves contextual understanding (Ji et al., 2021).

KGs are also widely applied in knowledge graph completion and link prediction, where the goal is to infer missing relationships between entities. Embedding models such as TransE and RotatE learn continuous representations that capture graph structure and enable prediction of unseen triples (Bordes et al., 2013; Sun et al., 2019).

In addition, KGs support reasoning and inference, including logical reasoning and rule-based traversal over graph structures. These capabilities are particularly useful in domains such as healthcare, finance, and recommendation systems, where explainability and structured decision-making are important (Hogan et al., 2021).

Finally, KGs are increasingly integrated with deep learning and large language models, enabling hybrid systems that combine symbolic reasoning with neural representations. This integration allows models to leverage both structured knowledge and unstructured text, improving performance across a wide range of tasks (Ji et al., 2021).

Overall, the versatility of Knowledge Graphs makes them a foundational component in modern intelligent systems, supporting a wide spectrum of applications that require structured knowledge representation and reasoning.

2.4 Knowledge Graphs in Recommender Systems

The integration of Knowledge Graphs represents a significant advancement in recommendation research, evolving from simple interaction matching to a sophisticated reasoning task performed over a Collaborative Knowledge Graph. A Collaborative Knowledge Graph unifies historical user-item interactions with heterogeneous item-attribute relationships into a singular, semantic latent space. Formally, a Collaborative Knowledge Graph is defined as $G = \{(h, r, t) \mid h, t \in E', r \in R'\}$, where the expanded entity set E' encompasses both users and items, and the relation set R' incorporates user behaviors such as interact alongside factual item descriptors (Wang et al., 2019a,b). By bridging these two domains, systems can distill collaborative signals from high-order connectivities, which are long-range dependencies that remain inaccessible to traditional matrix factorization models (Wang et al., 2019a). Current state-of-the-art Knowledge Graph aware recommendation models are fundamentally categorized into regularization-based, path-based, and propagation-based methodologies.

2.4.1 Regularization-based Methods

Regularization-based approaches, often referred to as embedding-based methods, leverage Knowledge Graph Embeddings to impose semantic constraints on the latent representations of items (Zhang et al., 2016). These models, exemplified by the Collaborative Knowledge Base Embedding framework, employ a multi-task learning objective:

$$L = L_{rec} + \lambda(L_{KG} + L_{reg}) \quad (2.3)$$

where L_{rec} optimizes for recommendation accuracy, typically via pairwise ranking or matrix factorization, and L_{KG} models the structural integrity of the Knowledge Graph

using translation-based distance functions such as TransR (Wang et al., 2017). In the TransR paradigm, each relation r is associated with a specific projection matrix M_r , which maps entities from the entity space to a relation-specific space. A factual triplet (h, r, t) is considered valid if the projected head entity plus the relation vector approximates the projected tail: $e_h M_r + e_r \approx e_t M_r$ (Zhang et al., 2016; Bordes et al., 2013).

This mechanism acts as a semantic filter, ensuring that items with shared attributes are proximal in the latent space. However, because these models primarily optimize for individual triplets, they effectively capture only direct, first-order relationships. High-order collaborative signals are modeled only implicitly through gradient propagation, which often proves insufficient for capturing complex, multi-hop dependencies (Wang et al., 2019a).

2.4.2 Path-based Reasoning Methods

In contrast to the implicit nature of regularization, path-based methods treat the Knowledge Graph as an explicit map of connectivity. Early iterations relied on meta-paths, which are hand-crafted connectivity patterns defined by domain experts. To overcome the rigidity and labor-intensiveness of meta-paths, the Knowledge-aware Path Recurrent Network was introduced to learn path semantics automatically via recurrent architectures (Wang et al., 2019b).

Knowledge-aware Path Recurrent Network extracts actual sequences of entities and relations $p = [e_1 \xrightarrow{r_1} e_2 \dots e_L]$ that connect users to items. It utilizes a Long Short-Term Memory network to encode these sequences, uniquely integrating entity types and relational semantics into the input vector x_l :

$$x_l = \phi(e_l \oplus e_{type} \oplus r_l) \quad (2.4)$$

By applying a weighted pooling or attention mechanism across all possible paths between a user and an item, the model can quantify the relative importance of different reasoning chains. The primary merit of path-based models is their inherent explainability, as they provide a transparent logical trace for each recommendation. Nevertheless, these models encounter significant scalability barriers, as the number of paths grows exponentially with the depth of the graph (Wang et al., 2019a; Ji et al., 2021).

2.4.3 Propagation-based Hybrid Methods

The current state-of-the-art, represented by the Knowledge Graph Attention Network, utilizes Graph Neural Networks to synthesize the efficiency of embedding methods with the multi-hop reasoning capability of path methods (Wang et al., 2019a). Knowl-

edge Graph Attention Network introduces an Attentive Embedding Propagation layer that recursively updates a node representation by aggregating messages from its local neighborhood.

The core innovation is the Knowledge-aware Attention mechanism, which determines the influence of a neighbor t on a head entity h conditioned on their specific relationship r :

$$\pi(h, r, t) = \frac{\exp(\text{LeakyReLU}(W_r e_t)^T \tanh(W_r e_h + e_r))}{\sum_{(h, r', t') \in N_h} \exp(\pi(h, r', t'))} \quad (2.5)$$

This attention flow allows the model to prioritize relevant semantic links during the information aggregation phase. By stacking L propagation layers, the model captures dependencies up to L -hops away, effectively automating the discovery of high-order collaborative signals in an end-to-end manner. Propagation-based models represent a robust solution for modern recommender systems, as they bypass the need for manual path extraction while maintaining a deep understanding of the global graph structure (Wang et al., 2019a; Ji et al., 2021).

CHAPTER 3: DATA

The validation of the proposed framework is conducted using two specialized datasets, each designed to address a distinct stage of the system’s lifecycle. The first dataset is dedicated to Skill Representation Learning, aiming to expose the embedding model to a wide array of domain-specific expressions. By training on diverse and specialized language across various educational contexts, the model enhances its ability to capture semantic similarities between skill descriptions. In contrast, the second dataset is designed for End-to-End Evaluation, providing a benchmark to measure the effectiveness of the complete recommendation pipeline—from job-candidate gap analysis to final course ranking. While the training dataset focuses on the precision of isolated skill mappings, the end-to-end dataset assesses the system’s practical utility in realistic scenarios.

3.1 Training Dataset for Skill Representation Learning

3.1.1 Data Description

To provide a consistent and high-quality reference space, skills are anchored to the ESCO (European Skills, Competences, Qualifications and Occupations) taxonomy, a multilingual classification system developed by the European Commission. ESCO provides a structured and standardized vocabulary of skills, occupations, and their relationships, serving as a common reference framework for the European labour market and education systems (European Commission, 2024). In this study, we use ESCO version v1.2 (released in May 2024), which contains 13,939 skills. The dataset is publicly available and can be downloaded from the official ESCO portal in multiple formats (European Commission, 2024). We focus specifically on the skill pillar of ESCO and apply filtering procedures to retain relevant and high-quality skill concepts. All data were downloaded from the official ESCO distribution interface, ensuring full reproducibility of the dataset construction process.

The dataset is structured as a triplet collection, where each record consists of three elements: a query (natural language expression of a skill), a positive (the corresponding correct ESCO skill description), and a hard negative sample (a semantically close but conceptually distinct skill of the query). This triplet design is grounded in metric learning theory, where models are trained to minimize the distance between query and positive while maximizing distance between query and negative. Each record also carries metadata: the ESCO URI (unique identifier), skill name, skill type (digital, transversal, etc.), alternative labels, source origin (synthetic or human-labeled), and

generation timestamp.

An illustrative data example is as follows:

```
{
  "query": "extract patterns and insights from large datasets",
  "positive": "Data analysis. The ability to examine and interpret...",
  "negative": "Data management. The ability to organize and store...",
  "metadata": {
    "skill_uri": "http://data.europa.eu/esco/skill/...",
    "skill_name": "data analysis",
    "skill_type": "digital",
    "alt_labels": ["data examination", "data interpretation"],
    "source": "auto_generated",
    "generated_at": "2025-03-23T10:30:00Z"
  }
}
```

The dataset comprises 41,806 auto-generated triplet samples across the ESCO skill taxonomy, split into 33,444 training samples and 8,362 test samples. The training set is reserved for model fine-tuning, while the test set includes both synthetic samples and 165 human-annotated mappings to enable external validation.

3.1.2 Data Preparation Process

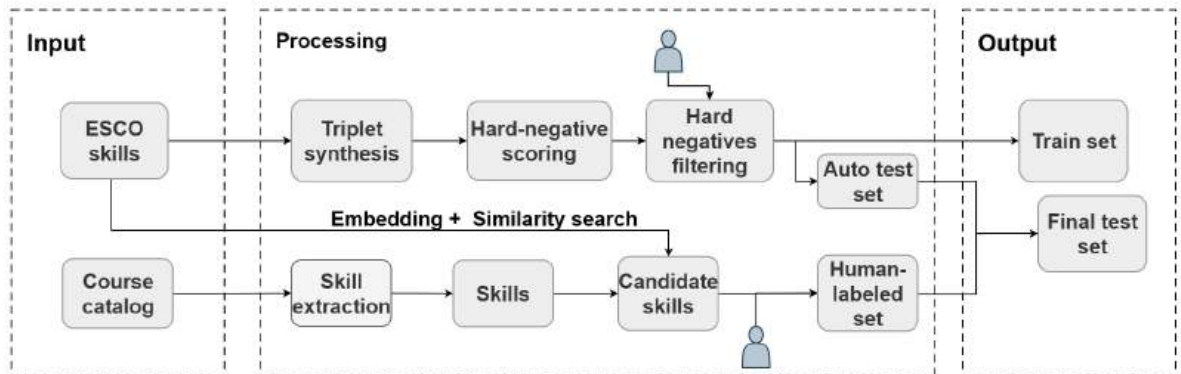


Figure 3.1 Training dataset preparation pipeline

The preparation process begins with systematic synthesis of training supervision from the ESCO taxonomy. For each ESCO skill, the process generates multiple query variations by prompting a large language model to create diverse phrasings that resemble how that skill might be expressed in course descriptions (detailed in Figure A.1). Simultaneously, hard negatives are synthesized by requesting the model to generate semantically related but fundamentally different skill descriptions.

Prior to training, the synthetic triplets undergo a rigorous quality control process

based on distributional analysis. For each sample, the system computes the *semantic margin* M , defined as the difference between the query-positive similarity and the query-negative similarity:

$$M = \text{sim}(Q, P) - \text{sim}(Q, N) \quad (3.1)$$

Where $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity between the vector representations of two text segments. Formally, given an encoder ϕ that maps text into a high-dimensional embedding space, the similarity is calculated as the normalized dot product:

$$\text{sim}(u, v) = \frac{\phi(u) \cdot \phi(v)}{\|\phi(u)\| \|\phi(v)\|} \quad (3.2)$$

This margin serves as the primary metric for assessing supervision quality:

- **Positive Margins** ($M > 0$): Indicate clear and reliable training signals where the current embedding space correctly prioritizes the positive mapping over the negative one.
- **Moderate Negative Margins** ($M < 0$, **small absolute value**): Suggest an insufficiency in the pre-trained model’s discriminative power. In these cases, the model fails to capture fine-grained domain nuances, making these samples highly valuable "hard negatives" for further fine-tuning to improve representation boundaries.
- **Extreme Negative Margins** ($M \ll 0$): Reveal potential "False Negatives", where the assigned negative sample is conceptually more relevant to the query than the intended positive. Such semantic inversions often stem from synthetic generation errors and introduce contradictory patterns that can degrade embedding quality.

To mitigate label noise, the pipeline implements a priority-based audit mechanism inspired by the positive-aware mining strategies in NV-Retriever (Moreira et al., 2024). The core logic follows the principle that negative samples with similarity scores exceeding a certain threshold relative to the positive score (e.g., $\text{sim}(Q, N) > \alpha \cdot \text{sim}(Q, P)$) are often mislabeled synthetic errors.

In this study, an audit was conducted on the top 1,000 samples exhibiting the most severe negative margins. A single annotator (the primary researcher) reviewed these triplets to verify if the "hard negative" actually described the same underlying skill as the query. The review criteria focused on identifying semantic overlaps such as: identical skills applied to different industries or job sectors, skills with nearly identical competencies but different specializations, or cases where the negative sample represented a broader hierarchical version of the positive skill.

To establish an empirical filtering threshold, the annotator reviewed samples in descending order of similarity until a sequence of 50 consecutive labels yielded no further "False Negatives" (i.e., all negatives were genuinely distinct skills, only sharing superficial phrasing). The similarity score at this stabilization point was then utilized as the global pruning threshold to strip the training pool of contradictory signals. Upon reaching this stabilization point, the resulting semantic margin was found to be $M = -0.152$, which was then utilized as the global pruning threshold to strip the training pool of contradictory signals. While the audit was performed by a single individual, consistency was maintained by strictly adhering to the ESCO taxonomy’s conceptual boundaries, ensuring the final dataset provides both the scalability required for deep representation learning and the empirical validity necessary for robust evaluation.

3.2 End-to-End Job-to-Course Recommendation Evaluation Dataset

3.2.1 Data Description

The end-to-end evaluation dataset is structured as a collection of **upskilling scenarios**. Each record represents a complete mapping from a professional requirement to a pedagogical solution, providing a benchmark for the entire recommendation pipeline.

Each data record consists of the following structured fields:

- **Input - Job Descriptions:** Contains the target role title, a list of required technical and soft skills, and minimum experience/education prerequisites.
- **Input - Candidate Profile:** Represents the learner’s current state, including their existing skills, current years of experience, and educational background.
- **Output - Course Suggestions:** A ranked list of courses retrieved from the curated academic catalog.

The final evaluation dataset comprises 100 job-candidate scenarios across a diverse range of industry domains, all of which were reserved exclusively for testing. This structured format allows for the calculation of standard Information Retrieval metrics by comparing the system’s ranked output against the human-provided relevance labels. To maintain the integrity of the evaluation, this test set was kept strictly separate from the synthetic triplet generation process used for domain-adaptive fine-tuning, thereby preventing any potential data leakage. By leveraging a non-parametric bootstrap approach on this high-quality dataset, we provide a robust estimation of the system’s performance, ensuring that our reported results are representative of the model’s true generalization capabilities in a specialized educational context.

3.2.2 Data Preparation Process

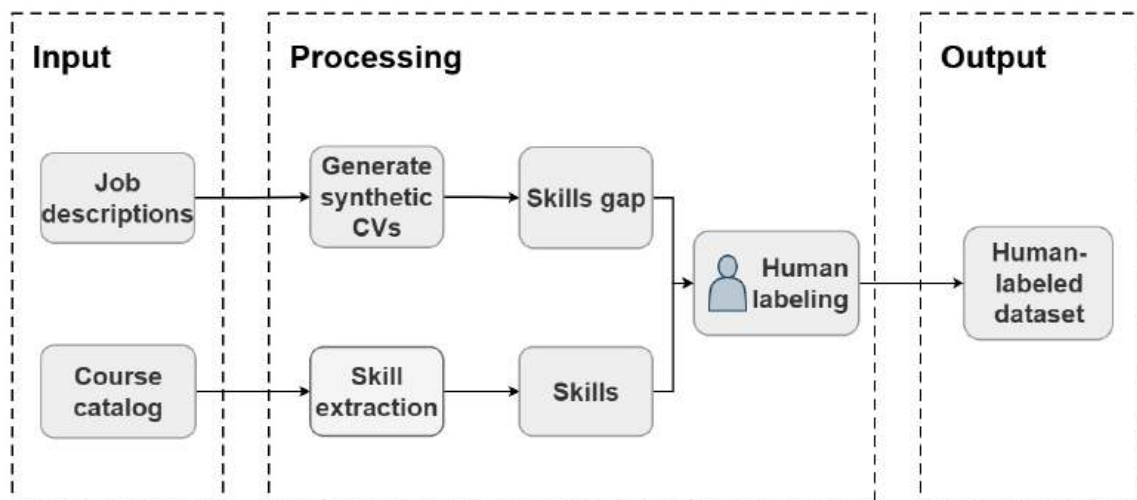


Figure 3.2 End to end dataset preparation pipeline

The pipeline initiates with the processing of authentic job descriptions to derive specific professional requirements. For each job posting, a synthetic candidate profile is generated with a controlled mismatch level, typically ranging from 60% to 80% alignment. This threshold is strategically enforced to simulate realistic career transitions and upskilling scenarios, avoiding the uninformative extremes of either perfect matches or complete irrelevance. The resulting candidate profile highlights explicit *Skills Gaps*, identifying the specific technical and soft competencies that the candidate lacks relative to the job’s demands, which subsequently serve as the primary query for recommendation. This synthesis step is governed by a structured prompt (detailed in Figure A.2).

Simultaneously, the supply-side branch focuses on transforming the raw course catalog into a structured, skill-centric knowledge base. A significant challenge in processing educational catalogs is that course descriptions often exhibit unbounded lengths and contain substantial amounts of non-pedagogical noise, such as instructor biographies, scheduling logistics, and administrative details. To mitigate this noise and ensure data consistency, the pipeline employs the Qwen3-14B large language model (LLM) and the prompt in Figure A.3 to perform skill extraction. The selection of Qwen3-14B is driven by its favorable trade-off between performance and deployment efficiency. With approximately 14.8 billion parameters, the model is lightweight enough to be self-hosted on commodity hardware while still maintaining strong reasoning and instruction-following capabilities (Yang et al., 2025). This enables full control over the inference pipeline and avoids reliance on external APIs. The model is implemented using the GGUF (GPT-Generated Unified Format) version with Q4_K_M quantization (LM-Kit, 2025). This specific 4-bit quantization scheme utilizes a hybrid bit-width approach, which sig-

nificantly reduces the VRAM requirements while maintaining a near-lossless level of perplexity compared to the original full-precision model. Furthermore, Qwen3-14B supports a native context length of up to 32K tokens, making it well-suited for processing long course descriptions without truncation (Yang et al., 2025). For the skill extraction task, we configure the model with a temperature of 0 to enforce deterministic and stable outputs, prioritizing consistency over diversity. This is particularly important for structured information extraction, where reproducibility and precision are critical. By leveraging the model’s advanced natural language understanding, the system distills verbose and unstructured curriculum metadata into a concise, classified set of skills. This extraction process not only filters out irrelevant information but also provides a standardized semantic representation of each learning module. Consequently, this refined output enables a robust and direct comparison between the core competencies offered by the courses and the previously identified candidate deficiencies.

The final stage of the pipeline is a manual labeling process conducted to create a high-quality evaluation set. Due to the large volume of courses, an initial filtering step was performed to remove clearly irrelevant options. Specifically, we applied a heuristic-based exclusion strategy to identify and remove non-instructional or administrative modules that do not contribute directly to granular skill acquisition. This involved filtering the course titles using a predefined list of keywords to exclude entries such as theses, internships, practicums, capstone projects, which effectively removed clearly irrelevant options, allowing the manual annotation effort to focus on core academic courses with substantial pedagogical content. The subsequent detailed labeling followed a strict set of rules to ensure accuracy. A course was marked as relevant only if it addressed at least one of the candidate’s identified skill gaps. For candidates with lower qualifications, foundational courses were prioritized over advanced ones to ensure a logical learning sequence. To improve efficiency, preference was given to courses that covered the maximum number of missing skills, reducing the total number of required learning modules. In cases where the catalog contained no suitable matches for a specific gap, no recommendation was assigned to maintain the integrity of the ground truth. This careful, rule-based approach resulted in a human-labeled set that serves as a reliable benchmark for evaluating the final recommendation system.

CHAPTER 4: METHODOLOGY

4.1 Domain-Adaptive Contrastive Learning

4.1.1 Model Selection

In this work, we utilize the Qwen/Qwen3-Embedding-0.6B checkpoint from Hugging Face as the embedding backbone. The model is part of the Qwen3 Embedding series, designed specifically for text embedding and retrieval tasks (Zhang et al., 2025).

The model choice is guided by the Massive Multilingual Text Embedding Benchmark (MMTEB) in February 2026 (Benchmark Version 2), a large-scale and comprehensive evaluation framework for text embedding models (Enevoldsen et al., 2025). MMTEB significantly extends previous benchmarks by covering 9 evaluation tasks across over 250 languages and 20 domains, including retrieval, semantic textual similarity, clustering, and classification tasks (Enevoldsen et al., 2025). This diversity makes it one of the most representative benchmarks for real-world embedding applications.

A key insight from MMTEB is that smaller embedding models (around 500M parameters) can achieve performance comparable to, or even surpass, much larger models on many tasks (Enevoldsen et al., 2025). This observation is particularly important for practical systems, where computational efficiency and latency are critical constraints.

Based on the MMTEB leaderboard¹, Qwen3-Embedding-0.6B demonstrates strong performance across multiple task categories while maintaining a relatively small model size. This makes it a suitable candidate for real-time systems that require both high-quality embeddings and low inference latency.

Importantly, the evaluation tasks in MMTEB closely align with our problem setting. In particular, semantic textual similarity and retrieval tasks require the model to accurately capture fine-grained semantic relationships between short text inputs. Since our application focuses on embedding short phrases such as skills and course descriptions, this capability is essential.

Qwen3-Embedding-0.6B has approximately 0.6 billion parameters, produces dense vector representations with a default embedding dimension of 1024, while also allowing flexible dimensionality ranging from 32 to 1024 depending on deployment requirements (Zhang et al., 2025).

Finally, the model is released under the Apache 2.0 license, enabling unrestricted academic and commercial use. It supports over 100 languages and is optimized for a

¹<https://huggingface.co/spaces/mteb/leaderboard>

wide range of downstream tasks, including semantic similarity, retrieval, clustering, and classification (Zhang et al., 2025).

4.1.2 Objective Function

To fine-tune the embedding model, we use the standard MultipleNegativesRankingLoss formulation with an explicit hard negative, which matches the training setup in our implementation.

Let a mini-batch contain N triplets (a_i, p_i, n_i) , where a_i is the anchor, p_i is the matched positive, and n_i is an explicit hard negative. For each anchor, all other positives in the batch are treated as in-batch negatives. Since the ESCO taxonomy consists of mutually exclusive skill concepts, the risk of encountering “soft false negatives”—where an in-batch negative term semantically overlaps with the anchor query—is inherently minimized.

The loss is defined as:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s(a_i, p_i))}{\exp(s(a_i, p_i)) + \exp(s(a_i, n_i)) + \sum_{\substack{j=1 \\ j \neq i}}^N \exp(s(a_i, p_j))} \quad (4.1)$$

where

- $s(x, y)$ denotes the similarity score produced by the encoder.

This objective is an InfoNCE-style contrastive loss. The positive pair (a_i, p_i) is encouraged to have the highest similarity among all candidates, while the explicit hard negative n_i and the in-batch negatives p_j for $j \neq i$ are pushed away.

If a batch does not contain explicit negatives, the term $\exp(s(a_i, n_i))$ is simply omitted, and the formula reduces to the standard Multiple Negatives Ranking Loss.

The in-batch negatives provide a global separation signal across the batch, while the explicit hard negative sharpens the local decision boundary between semantically similar but incorrect candidates. This is why the loss is well suited for fine-grained skill discrimination.

4.2 Graph-based Recommendation system

This section describes the design and implementation of the Graph-based Recommendation System, which leverages structured knowledge to transform raw course data into actionable, sequential learning paths. The system bridges the gap between semantic representation and pedagogical logic by utilizing a Knowledge Graph (KG) for reasoning.

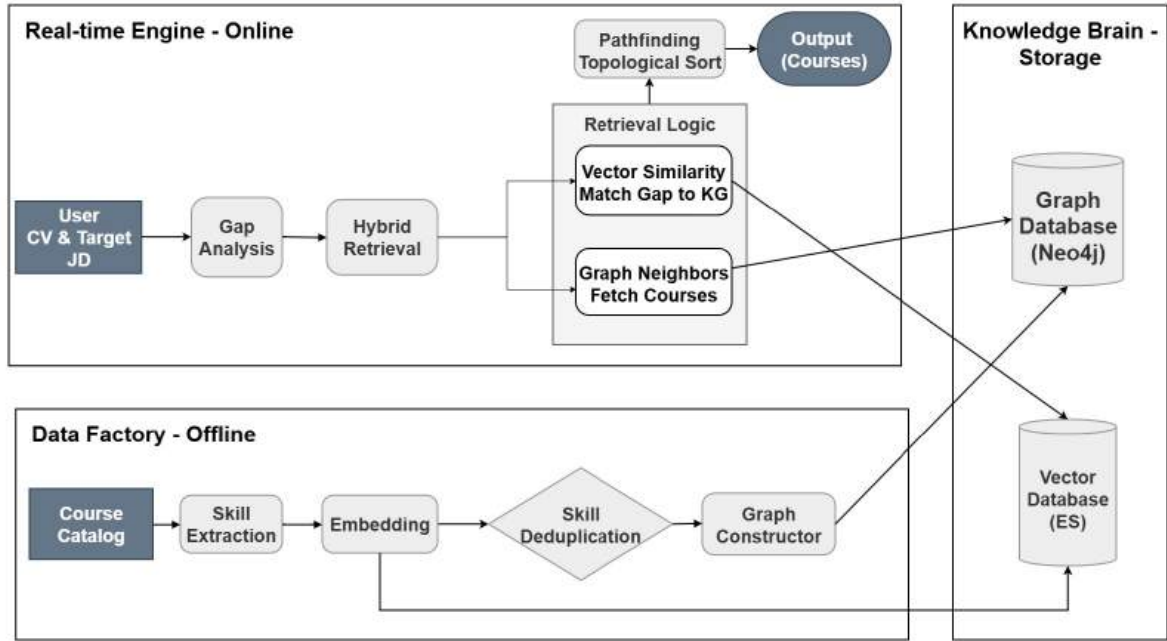


Figure 4.1 End to end recommendation pipeline

4.2.1 Schema and Ontology Design

To ensure the system can reason about learning sequences and provide explainable suggestions, we define a formal ontology consisting of four primary entities and two core relationship types.

4.2.1.1 Core Entities

- **Course:** The fundamental learning unit, characterized by its curriculum metadata and the specific competencies it imparts to learners.
- **Skill:** A canonical competency node. Unlike raw text labels, these nodes represent standardized skills used for cross-domain retrieval and reasoning.
- **Candidate Profile:** A dynamic entity representing the learner's current state, derived from skills pre-extracted from their CV.
- **Job Description:** The target state for the learner, consisting of a set of skills pre-extracted from a target Job Description (JD).

4.2.1.2 Semantic Relationships

The graph's intelligence is encoded in its edges:

- **TEACHES (Course → Skill):** Defines the positive outcomes of a learning unit. This relationship is used to identify which courses can resolve specific skill gaps.

- **REQUIRES (Course $\rightarrow$ Skill):** Establishes the prerequisite constraints. This is critical for generating logically ordered paths where foundational skills are acquired before advanced modules.

This ontology supports *explainability* by allowing the system to justify a recommendation (e.g., "Course A is recommended because it teaches Skill X, which is a required gap in your profile").

4.2.2 Skill Canonicalization and Graph Construction

A significant challenge in skill-based systems is the high degree of lexical and semantic variation in how skills are described. We implement an offline **Data Factory** pipeline to construct a clean, deduplicated Knowledge Graph.

4.2.2.1 Two-Stage Deduplication Pipeline

To convert raw extracted skills into canonical nodes, we employ a two-stage process:

- **Stage 1 (Lexical):** Fuzzy matching with a similarity threshold of 0.9 is used to merge minor variations, such as typos and formatting differences (e.g., "Python3" vs. "Python 3").
- **Stage 2 (Semantic):** Utilizing the domain-adaptive embedding model described in Section 4.1, we compute cosine similarity between skill vectors. Skills with a similarity score exceeding a configurable threshold ($\tau_{dedup} = 0.9$) are merged into a single canonical node, with original terms preserved as aliases for traceability.

4.2.2.2 Storage and Indexing

The resulting structure is persisted in a hybrid storage architecture:

- **Neo4j:** A graph database used to store the relational structure, enabling efficient multi-hop traversal for prerequisite checking.
- **Elasticsearch:** A search and analytics engine utilized for its vector indexing capabilities. It stores embeddings of canonical skills to facilitate rapid semantic retrieval.

4.2.3 Automated Skill-Gap Identification

The recommendation process begins with the gap analysis module, which identifies the discrepancy between the user's current capabilities and the job's demands.

Given a set of target skills S_{JD} and a set of existing skills S_{CV} , we identify the missing skills ΔS using a semantic-aware decision rule. For each target skill $s \in S_{JD}$, we

calculate its maximum similarity to the user's existing skill set:

$$\text{IsGap}(s) = \begin{cases} \text{True} & \text{if } \max_{s' \in S_{CV}}(\text{sim}(e_s, e_{s'})) < \tau_{gap} \\ \text{False} & \text{otherwise} \end{cases} \quad (4.2)$$

where e_s is the embedding vector and τ_{gap} is a configurable threshold. This approach prevents the system from suggesting courses for skills that the user already possesses but are phrased differently in their CV. It is important to clarify that while this automated identification module is designed for the system's practical deployment to process raw resumes, it was intentionally bypassed during the experimental evaluation (Chapter 5). To ensure a controlled experimental environment and maintain high-quality supervision signals, the benchmarks were conducted using pre-identified skill gaps from the synthesis dataset. This setup ensures that the reported metrics strictly reflect the retrieval and reasoning performance of the recommendation engine rather than the potential noise introduced by initial skill extraction from CVs.

4.2.4 Hybrid Retrieval and Sequential Path Reasoning

The final stage of the methodology involves mapping the identified gaps to a coherent learning roadmap. This process transitions from semantic matching to structural graph reasoning.

4.2.4.1 Multi-stage Retrieval

The system first performs a hybrid retrieval:

- **Vector Matching:** Map each gap term in ΔS to its nearest canonical skill nodes in the graph via vector search.
- **Neighbor Fetching:** Query the graph database to retrieve all courses that have a TEACHES relationship with these target canonical skills.

4.2.4.2 Topological Course Sequencing

Once a set of candidate courses is retrieved, they are initially unordered. To transform this set into a Personalized Learning Path, the system employs a Sequential Reasoning mechanism based on the Knowledge Graph's structure. First, the system inspects the REQUIRES edges to establish dependencies; if a course C_2 requires a skill taught by course C_1 , a directed dependency $C_1 \rightarrow C_2$ is enforced. To ensure the robustness of the sequencing process, a cycle detection step using Depth-First Search is performed to identify any mutual dependencies or logical loops that may exist in the source data. When a cycle is detected, the system evaluates the dependency count for each course involved in the loop. This count is defined as the number of a course's REQUIRED skills that are

provided by other courses within the candidate set. The course with the lowest number of incoming dependencies is identified as the most foundational within that specific context. To break the cycle, the system removes the edge pointing to this foundational course from within the loop, thereby designating it as the starting point. This ensures the resulting structure is a strictly Directed Acyclic Graph (DAG). Finally, a Topological Sort is applied to the DAG, producing a pedagogically sound recommendation sequence that prioritizes foundational content and respects the inherent learning order.

CHAPTER 5: EXPERIMENTAL RESULTS

5.1 Experimental setup

5.1.1 *Embedding fine-tuning setup*

5.1.1.1 *Implementation Environment*

The fine-tuning process is implemented using the Hugging Face Transformers and PEFT libraries. All experiments are conducted on a single NVIDIA GeForce RTX 5090 GPU (32GB VRAM). To maximize the available memory and accelerate computation, we utilize mixed-precision training with bfloat16 (bf16) and enable gradient checkpointing, which allows for larger effective batch sizes by recomputing intermediate activations during the backward pass.

5.1.1.2 *Training Configuration*

The selection of hyperparameters, summarized in Table 5.1, aims to balance fine-grained representation quality with the risk of overfitting in a domain-specific context.

We set the maximum sequence length to 512 tokens. While skill phrases are typically short, this capacity ensures the model can process extended course descriptions without information loss. We employ an effective batch size of 64, achieved via gradient accumulation over 4 steps. In contrastive learning, batch size is a critical performance driver; since we utilize in-batch negatives, a larger batch provides $N - 1$ diverse negative samples for each anchor, significantly enhancing the discriminative power of the learned embedding space (Gao et al., 2021).

The model is optimized using a learning rate of 2×10^{-5} to prevent the collapse of pre-trained weights during domain adaptation. A linear warm-up ratio of 10% is applied to stabilize the initial gradients. This conservative optimization strategy ensures that the model adapts to the specific semantics of professional skills and academic courses without losing general linguistic knowledge.

Following the InfoNCE framework, we adopt an in-batch negative sampling strategy. Every positive pair within a mini-batch serves as an implicit negative for all other pairs in the same batch. This approach provides a computationally "free" source of diverse negatives. By training the model to distinguish a specific skill from a wide range of other domain-related skills in the same batch, we force the model to capture fine-grained semantic boundaries rather than relying on coarse linguistic patterns.

The training process is regulated by an early stopping mechanism with a patience of 3 epochs and a minimum improvement threshold of 0.001 on the validation loss. This

Table 5.1 Hyperparameter settings for embedding fine-tuning.

Hyperparameter	Value
Base Model	Qwen3 Embedding 0.6B
Max Sequence Length	512 tokens
Per-device Batch Size	16
Gradient Accumulation Steps	4
Effective Batch Size	64
Learning Rate	2×10^{-5}
Learning Rate Scheduler	Linear with Warm-up
Warm-up Ratio	10%
Max Epochs	5
Mixed Precision	bf16
Early Stopping Patience	3 epochs
Train/Validation Split	90% / 10%

prevents the model from over-specializing on the synthetic training triplets. The final model is selected based on the checkpoint achieving the lowest validation loss to ensure maximum generalization across unseen course catalogs.

5.2 Evaluation metrics

To rigorously assess the proposed system, we employ a two-tier evaluation framework focusing on: Skill Alignment (the quality of the domain-adaptive embeddings) and Course Recommendation (the performance of the end-to-end pipeline). Both tiers are evaluated using a shared suite of ranking metrics, including Hit Rate@k, MRR, and NDCG. While all metrics are applied to the course tier, Recall is omitted for the skill tier because each query has only one ground-truth target (making it identical to Hit Rate), whereas it is essential for the course tier to capture the retrieval of multiple relevant items per recommendation. The definitions and contextual applications of these metrics are detailed below.

5.2.1 Hit Rate@k (HR@k)

Hit Rate@k (HR@k) measures the proportion of queries for which the ground-truth positive sample is successfully retrieved within the top k ranked results.

- **Calculation:** For each query, a "hit" is recorded as a binary indicator (1 if successful, 0 otherwise) if the rank of the positive item r_{pos} satisfies the condition $r_{pos} \leq k$. The final HR@k value is the average of these indicators across the entire test set.
- **Aspect:** It evaluates the model's effectiveness in identifying the correct semantic alignment. HR@1 is the most stringent metric, requiring the model to prioritize the exact match as the top candidate. Higher values of k (e.g., HR@3 or HR@5) provide insights into the model's robustness in capturing semantic proximity, even if the exact match is not the first result.

5.2.2 Mean Reciprocal Rank (MRR@k)

MRR@k accounts for the specific position of the first relevant item within the top k results. If the positive sample appears beyond rank k , its reciprocal rank is treated as 0.

- **Calculation:** $MRR@k = \frac{1}{|Q|} \sum_{i=1}^{|Q|} f(rank_i)$, where $f(rank_i) = \frac{1}{rank_i}$ if $rank_i \leq k$, and $f(rank_i) = 0$ otherwise.
- **Evaluation Aspect:** It evaluates the model's efficiency in pushing the correct semantic match to the top of the list, while strictly penalizing failures to retrieve the target within the user's immediate attention span (the top k results).

5.2.3 Recall@k

Recall@k calculates the ratio of relevant courses successfully retrieved in the top k results compared to the total number of relevant courses in the ground truth.

- **Calculation:** $Recall@K = \frac{|Relevant\ Items \cap Retrieved\ Items@K|}{|Total\ Relevant\ Items|}$.
- **Evaluation Aspect:** It measures the "completeness" of the recommendation. A high Recall@10 ensures that the system does not miss potential courses that could cover the user's skill gaps.

5.2.4 Normalized Discounted Cumulative Gain (NDCG@k)

NDCG evaluates the ranking quality by considering the position of relevant items, where items appearing earlier in the list are given higher weights.

- **Calculation:** $NDCG@K = \frac{DCG@K}{IDCG@K}$, where DCG is the sum of relevance scores discounted by the logarithm of their position.
- **Evaluation Aspect:** Because it is normalized against an ideal ranking (IDCG), it effectively handles cases where the number of relevant items varies across different users, providing a fair assessment of ranking order.

5.2.5 Mean Average Precision (MAP)

MAP provides a single-figure measure of quality across all levels of the recommendation list.

- **Calculation:** For a given query, the Average Precision (AP) is defined as the mean of the precision scores calculated at each position where a relevant item is retrieved. In the specific context of our evaluation, where a query might contain only a single relevant item ($R = 1$) located at rank k , the AP formula simplifies as follows:

$$AP = \frac{1}{R} \sum_{i=1}^n P(i) \times rel(i) = \frac{1}{1} \times (P(k) \times 1) = \frac{1}{k} \quad (5.1)$$

In skill tier, since the precision at rank k is exactly $1/k$, the AP score for that query becomes identical to its Reciprocal Rank (RR). Consequently, the aggregate MAP across the entire test set yields the same result as the Mean Reciprocal Rank (MRR), reflecting the system’s efficiency in placing the target item as high as possible in the recommendations.

- **Evaluation Aspect:** MAP is particularly well-suited for course recommendations because it effectively handles variations in ground-truth set sizes. Whether a learner requires a long sequence of courses or just a single module, MAP evaluates the ranking quality across the entire retrieved list. Unlike fixed-cutoff metrics, it rewards the system for placing all relevant items higher in the results, making it a reliable indicator of the overall effectiveness of the pedagogical roadmap.

5.2.6 Statistical Robustness and Confidence Intervals

Given the specialized nature of the domain-specific evaluation dataset ($n = 100$), we prioritized label quality over quantity. To ensure that the performance improvements are statistically robust and not artifacts of sampling variance, we employed a non-parametric percentile bootstrap procedure. For each model configuration, we performed 1,000 resampling iterations with replacement to estimate the 95% Confidence Intervals (CI) for all primary metrics. This approach provides a measure of stability for our point estimates, allowing for a more rigorous comparison between the baseline and the proposed KG-augmented framework.

5.3 Results & Discussion

5.3.1 Embedding Model Fine-tuning

The experimental results, summarized in Table 5.2, demonstrate a consistent and substantial improvement across all evaluated metrics following the domain-adaptive fine-tuning process. The fine-tuned model (Post-training) consistently outperforms the base backbone model (Pre-training) at every threshold. This uniform growth across Hit Rate, MRR, NDCG, and MAP validates that the contrastive learning strategy using triplet samples and in-batch negative training effectively adapts the general-purpose embedding space to the specific semantic requirements of the labor market and educational domains.

k	Hit Rate@k		MRR@k		NDCG@k		MAP@k	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post
@1	0.6415	0.7272	0.6415	0.7272	0.6415	0.7272	0.6415	0.7272
@3	0.7986	0.8936	0.7119	0.8023	0.7342	0.8258	0.7119	0.8023
@5	0.8487	0.9341	0.7234	0.8116	0.7549	0.8426	0.7234	0.8116
@10	0.8896	0.9673	0.7290	0.8162	0.7683	0.8535	0.7290	0.8162

Table 5.2 Performance comparison of the Embedding Model before and after fine-tuning. Pre: Pre-training (Backbone); Post: Post-training.

The most significant improvement is observed in Hit Rate@1, which increased from 0.6415 to 0.7272 (a relative improvement of approximately 13.36%). In the context of skill-to-course matching, HR@1 is a high-precision metric that measures the model’s ability to place the exact canonical match at the very top of the list. Such an improvement indicates that the fine-tuning process has successfully sharpened the model’s discriminative boundaries, allowing it to distinguish between nuanced technical terms that general-purpose models often conflate.

The model’s robustness is further highlighted by the Hit Rate@3 and Hit Rate@10 scores, which reached 0.8936 and 0.9673, respectively. Achieving a Hit Rate@3 of nearly 90% implies that the correct skill representation is captured within the top three candidates for the vast majority of queries. This high level of coverage is critical for the stability of the subsequent Hybrid Retrieval module, as it ensures that the "skills gap" identified from a candidate’s CV is almost certainly mapped to the correct entities in the Knowledge Graph. By the time k=10, the model achieves a competitive Hit Rate of 0.9673, effectively eliminating the risk of missing relevant foundational skills.

Regarding ranking quality, the improvements in $\text{MRR}@k$ and $\text{NDCG}@k$ reflect a superior ability to push relevant items toward the top of the recommendation list. $\text{NDCG}@3$ saw a remarkable absolute gain of 0.0916, rising from 0.7342 to 0.8258. Since NDCG and MRR place higher weights on items at the first few positions, these gains prove that the fine-tuning did not merely include the correct items in the results but significantly optimized their relative ranking.

Notably, the MAP (Mean Average Precision) values are identical to the MRR scores across all levels. This alignment is a direct result of the evaluation dataset structure, where each query is mapped to a single ground-truth positive. The simultaneous rise in these metrics confirms that the domain-adaptive embedding model provides a highly reliable foundation for the pedagogical reasoning engine, ensuring that foundational courses are prioritized with high confidence.

5.3.2 Recommendation System

The evaluation results are summarized in Table 5.3. The most prominent finding is the massive performance gap between the standard embedding search (pre-trained)—which serves as the baseline—and the KG-augmented recommendation (fine-tuned). At the most challenging evaluation level $k=1$, the proposed system achieved an $\text{HR}@1$ of 0.590, representing a 126.9% relative improvement over the baseline’s 0.260. As the recommendation window expands to $k=5$, the system reaches a near-optimal Hit Rate of 0.920, ensuring that a relevant learning solution is almost always present in the top-tier results.

Although some overlap in the Hit Rate confidence intervals is observed at $k=5$ due to performance saturation (where most models successfully retrieve at least one relevant course), the consistent superiority of the KG-augmented model in NDCG and MRR demonstrates its ability to push the most logically appropriate courses to the top-tier positions.

Table 5.3 Comprehensive performance overview with 95% Confidence Intervals
($n = 100$). CI are reported in brackets below each mean value.

k	Method	Hit Rate	Recall	NDCG	MRR	MAP
@1	Standard (Pre-trained)	0.260 [0.180, 0.350]	0.089 [0.058, 0.126]	0.260 [0.180, 0.350]	0.260 [0.180, 0.350]	0.260 [0.180, 0.350]
	Standard (Fine-tuned)	0.540 [0.440, 0.640]	0.231 [0.179, 0.291]	0.540 [0.440, 0.640]	0.540 [0.440, 0.640]	0.540 [0.440, 0.640]
	KG-augmented (Pre-trained)	0.570 [0.470, 0.670]	0.215 [0.167, 0.265]	0.570 [0.470, 0.670]	0.570 [0.470, 0.670]	0.570 [0.470, 0.670]
	KG-augmented (Fine-tuned)	0.590 [0.490, 0.690]	0.222 [0.175, 0.276]	0.590 [0.490, 0.690]	0.590 [0.490, 0.690]	0.590 [0.490, 0.690]
@3	Standard (Pre-trained)	0.560 [0.460, 0.660]	0.295 [0.232, 0.360]	0.309 [0.248, 0.374]	0.388 [0.312, 0.472]	0.243 [0.191, 0.301]
	Standard (Fine-tuned)	0.830 [0.750, 0.900]	0.490 [0.419, 0.558]	0.529 [0.462, 0.590]	0.668 [0.595, 0.742]	0.447 [0.383, 0.511]
	KG-augmented (Pre-trained)	0.820 [0.740, 0.890]	0.401 [0.348, 0.461]	0.479 [0.419, 0.537]	0.685 [0.603, 0.762]	0.386 [0.330, 0.442]
	KG-augmented (Fine-tuned)	0.870 [0.800, 0.930]	0.489 [0.436, 0.552]	0.565 [0.505, 0.627]	0.718 [0.652, 0.788]	0.480 [0.419, 0.544]
@5	Standard (Pre-trained)	0.650 [0.550, 0.740]	0.385 [0.312, 0.451]	0.333 [0.271, 0.390]	0.409 [0.332, 0.491]	0.249 [0.199, 0.298]
	Standard (Fine-tuned)	0.900 [0.840, 0.960]	0.604 [0.535, 0.671]	0.565 [0.502, 0.627]	0.685 [0.612, 0.754]	0.468 [0.405, 0.530]
	KG-augmented (Pre-trained)	0.860 [0.790, 0.920]	0.509 [0.444, 0.574]	0.498 [0.440, 0.553]	0.694 [0.615, 0.768]	0.387 [0.334, 0.442]
	KG-augmented (Fine-tuned)	0.920 [0.870, 0.970]	0.589 [0.534, 0.651]	0.572 [0.519, 0.629]	0.730 [0.664, 0.798]	0.463 [0.410, 0.522]

5.3.2.1 Impact of Domain-Adaptive Fine-tuning

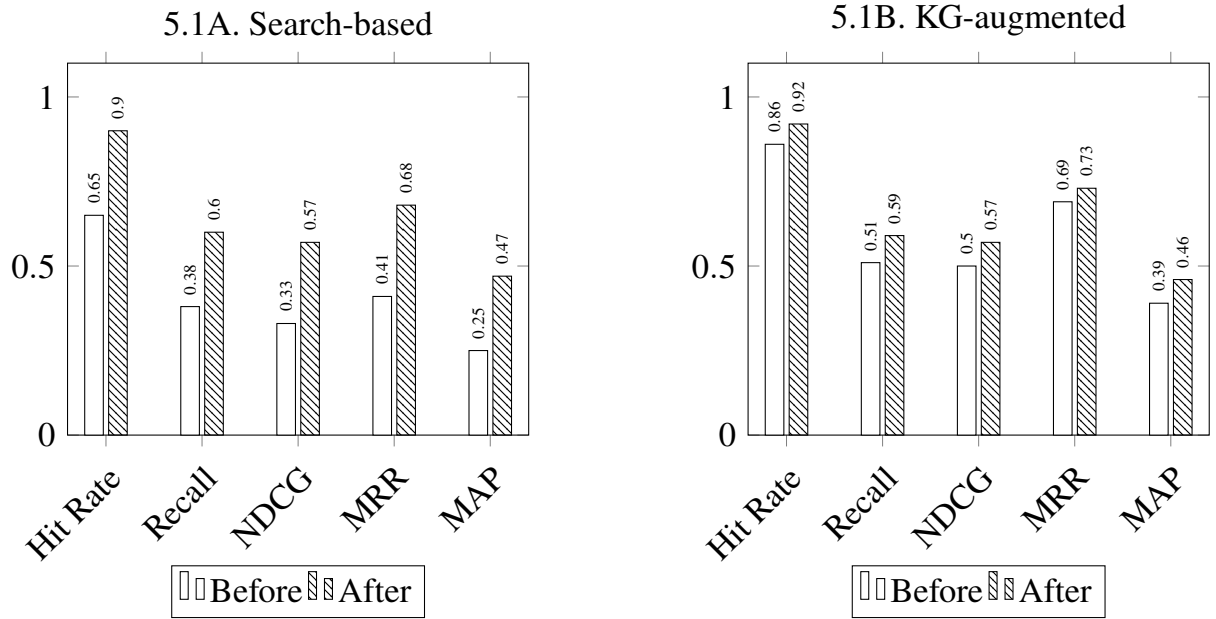


Figure 5.1 Performance enhancement attributed to Domain-Adaptive Fine-tuning at $k = 5$.

The transition from pre-trained to fine-tuned embeddings, as visualized in Figures 5.1, marks a fundamental shift in the system’s ability to interpret professional competencies. In Chart 5.1A, the standard embedding search shows a dramatic surge across all metrics post-training; for instance, Hit Rate@5 jumped from 0.65 to 0.90, and Recall@5 increased from 0.385 to 0.604.

This improvement validates the effectiveness of the Domain-Adaptive Contrastive Learning strategy described in Chapter 4. By utilizing LLM-generated synthetic triplets and, more importantly, challenging "Hard Negatives," the model successfully sharpened its discriminative boundaries. In the context of course recommendation, this means the system no longer merely relies on surface-level keyword matching but can distinguish between conceptually similar yet functionally distinct skills. The robust increase in MAP and NDCG in both Standard and KG-augmented versions proves that fine-tuning creates a more coherent semantic space where relevant educational content is brought significantly closer to the identified skill gaps.

As observed in Figure 5.1B, the KG-augmented configuration also exhibits steady performance gains post-fine-tuning, although the margin of improvement is smaller compared to its search-based counterpart. This phenomenon is attributed to the high initial baseline achieved by the KG-augmented system even in its pre-trained state. This suggests that the Knowledge Graph effectively compensates for the semantic limitations of general-purpose embeddings by guiding the retrieval process toward logically relevant nodes. In this scenario, fine-tuning serves as a final layer of semantic refinement,

providing the high-fidelity matching needed to push an already logically sound system toward highly effective performance, reaching a peak Hit Rate of 0.92 and MRR of 0.73.

5.3.2.2 Impact of Knowledge Graph Reasoning

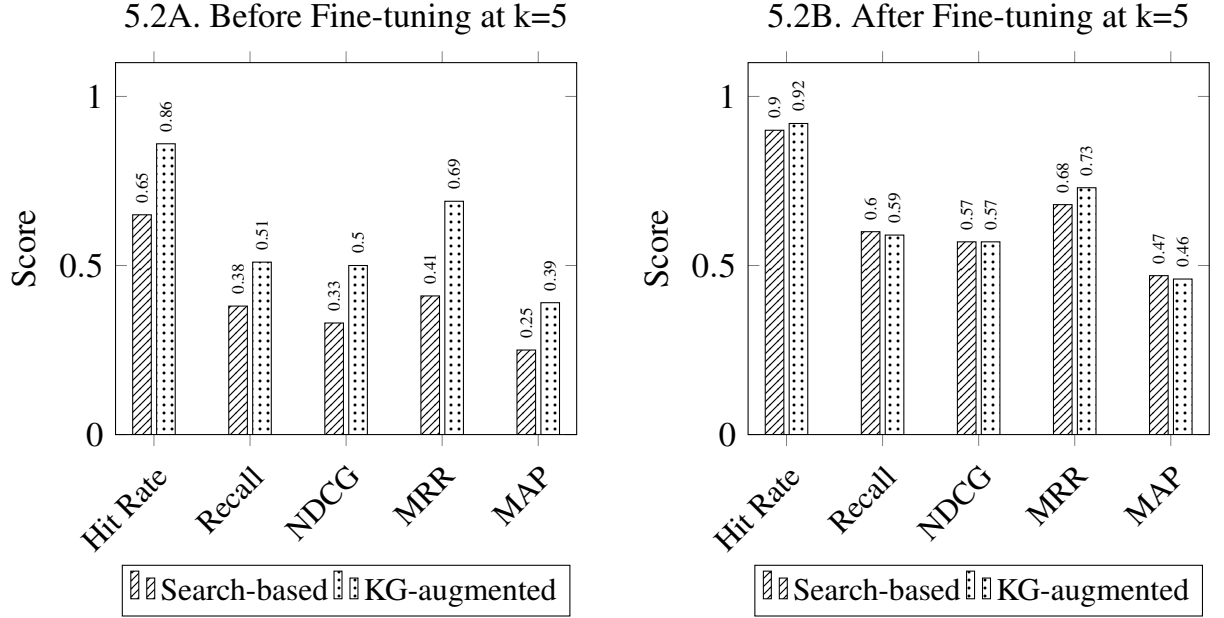


Figure 5.2 Performance enhancement attributed to Knowledge Graph reasoning at $k = 5$.

While fine-tuning optimizes the "content" layer of the recommendation, the integration of Knowledge Graph (KG) reasoning provides the "logical" layer. Chart 5.2A reveals a compelling insight: even before any fine-tuning was applied, the KG-augmented Recommendation significantly outperformed the Standard baseline in ranking metrics. Specifically, the MRR@5 reached 0.69 for the KG version compared to only 0.41 for the Search-based version.

This performance leap is a direct result of the TEACHES and REQUIRES relational edges. Unlike standard vector search, which is prone to "semantic drift" (recommending courses that are textually similar but pedagogically irrelevant), the KG-augmented system enforces a canonical mapping to skill nodes. The high MRR and NDCG scores across all k levels in Table 5.3 indicate that the Topological Course Sequencing logic effectively prioritizes foundational content. By respecting prerequisite constraints, the system ensures that the most logically appropriate courses for a beginner's career path are pushed to the top-1 position.

However, a more nuanced observation can be made when comparing the Standard (Fine-tuned) and KG-augmented (Fine-tuned) models at $k = 5$. The integration of the

Knowledge Graph leads to a slight trade-off in Recall@5 (from 0.604 to 0.589) and a marginal decline in MAP (from 0.468 to 0.463). This trade-off suggests that while the KG-based reasoning is highly effective at retrieving relevant courses that semantic search alone might miss, the topological re-ranking might slightly shift the primary semantic matches further down the list to accommodate prerequisite or logically related nodes.

CHAPTER 6: DISCUSSION

6.1 Interpretations of Results

The successful evaluation of our framework demonstrates the effectiveness of the designed framework. This section analyzes the experimental findings and explains how these techniques contribute to both theoretical knowledge and practical applications. We focus on the synergy between semantic alignment and pedagogical logic, while also discussing the trade-offs between performance and system complexity. Finally, we address the limitations of the current artifact and suggest directions for future research.

6.2 Contributions to the Literature

This research makes several distinct contributions to the field of educational recommendation systems.

- **Methodology for Data-Scarce Domains:** We demonstrate a robust pipeline for utilizing Large Language Models (LLMs) to synthesize high-quality triplet datasets. By specifically focusing on "Hard Negatives," we provide a blueprint for domain adaptation in specialized fields where human-labeled data is prohibitively expensive.
- **A Domain-Adapted Model via Customized Datasets:** This study also contributes a specialized embedding model, specifically fine-tuned using a high-quality, synthetic triplet dataset tailored for the educational domain.
- **Knowledge Graph based Recommendation System:** Finally, we instantiate a structured Knowledge Graph built upon a formal ontology with clear and strictly defined relationships. By encoding logical dependencies, the framework ensures that recommendations move beyond simple keyword matching to provide coherent, executable learning sequences that respect inherent prerequisite constraints.

6.3 Practical Implications: The Performance-Complexity Trade-off

From a practitioner's perspective, the results reveal a critical decision point regarding system architecture. While the KG-augmented (Fine-tuned) model achieved the highest benchmark scores (0.92 Hit Rate, 0.73 MRR), the margin of improvement over the Standard Embedding Search (with Fine-tuned model)—which reached a 0.90 Hit Rate—is relatively slim.

For organizations seeking to maximize every percentage point of accuracy and provide high explainability, the full KG-augmented approach is the superior choice. However,

for large-scale production environments where system latency, maintenance costs, and database complexity are primary concerns, the Fine-tuned Embedding Search alone offers an exceptionally attractive "middle ground." It maintains over 97% of the peak Hit Rate and 93% of the MRR achieved by the full KG-augmented system with significantly lower computational overhead, suggesting that in many real-world applications, a well-trained domain-adaptive model may suffice without the need for complex graph traversals.

6.4 Limitations and Future Work

Despite the high performance observed, several constraints remain that warrant further investigation. Primarily, the reliance on LLM-generated triplets for domain adaptation introduces risks of synthetic data bias. While rigorous quality control was implemented, such synthetic distributions may not fully capture the intricate semantic nuances of real-world academic curricula. Furthermore, the system's integrity is heavily dependent on the LLM's extraction capabilities for Knowledge Graph construction; any errors or "hallucinations" during the entity and relationship extraction phase can propagate through the pedagogical reasoning engine, leading to logically flawed recommendations.

Regarding evaluation, the current study is limited by a relatively small end-to-end (E2E) validation set, which may not represent the full diversity of learner profiles and career paths. The data auditing and threshold-setting process were also conducted by a single primary annotator. While internal consistency was maintained by adhering to the ESCO taxonomy, the absence of multiple independent coders prevents the calculation of robust inter-annotator agreement (IAA) metrics, potentially introducing subjective bias. Furthermore, the stabilization-based sampling—which relied on a heuristic of 50 consecutive correct labels to determine the pruning threshold—represents an empirical shortcut. Although this approach is practical for managing large-scale synthetic data, it may fail to identify "long-tail" errors or rare semantic inversions that exist beyond the stabilization point.

Moreover, there is a lack of a formal user study; while offline metrics like Recall and MAP provide quantitative evidence of performance, they cannot assess qualitative factors such as the perceived helpfulness, user satisfaction, or the actual learning impact of the recommended roadmaps. Additionally, this research did not include an ablation study for specific graph constraints. Consequently, the individual contributions of different relational edges, such as TEACHES versus REQUIRES, have not been isolated, leaving the specific drivers of the performance gain partially unquantified.

Finally, the system faces a 'cold-start' challenge for emerging technologies not yet in the ontology and potential scalability bottlenecks in the sequential pathfinding logic as the course catalog expands toward the million-scale.

Future research will focus on several key directions:

- **Human-in-the-loop Evaluation:** Conducting comprehensive user studies to validate the pedagogical coherence of the recommendations and align offline metrics with real-world student success.
- **Ablation Analysis:** Performing detailed experiments to decompose the KG architecture and quantify the exact impact of each logical constraint (e.g., prerequisite vs. topical alignment).
- **Hybrid Knowledge Extraction:** Integrating symbolic NLP with LLMs to improve the robustness of skill extraction and reduce dependency on purely generative models.
- **Dynamic Learning:** Utilizing Reinforcement Learning to adjust learning paths based on real-time feedback, and expanding to multimodal inputs (video/audio) to extract skills from diverse educational formats.

CHAPTER 7: CONCLUSION

This research presented a comprehensive framework for enhancing course recommendations by integrating Domain-Adaptive Contrastive Learning with Graph-based Reasoning. By addressing the dual challenges of labeled data scarcity and the lack of pedagogical logic in traditional recommendation systems, the study demonstrated that a synergy between deep semantic understanding and structural reasoning is essential for creating effective personalized learning paths.

The proposed solution was built upon two foundational pillars. First, to overcome the limitations of general-purpose embeddings, we developed a domain-adaptive model fine-tuned on a customized triplet dataset. The use of Large Language Models to generate challenging “hard negatives” proved highly effective, allowing the model to distinguish between nuanced technical competencies and significantly reducing semantic drift. Second, we implemented a structured Knowledge Graph that encodes explicit pedagogical relationships through *TEACHES* and *REQUIRES* edges. This logic-driven layer transformed the recommendation process from simple similarity matching into a sequential pathfinding engine that respects inherent prerequisite constraints.

The experimental results provided robust evidence of the framework’s superiority. The proposed system achieved Hit Rate@1 of 0.590 and a near-optimal Hit Rate@5 of 0.920, representing a massive leap over pre-trained baselines. Specifically, the high MRR and NDCG scores confirmed that the Knowledge Graph reasoning effectively prioritizes foundational courses, ensuring that the generated roadmaps are not only relevant but also practically feasible for the learner.

In terms of contributions, this research provides a scalable methodology for domain adaptation in data-scarce environments and a hybrid architecture that bridges the gap between “soft” vector similarity and “hard” logical constraints. From a practical standpoint, we identified a performance-complexity trade-off: while the full KG-augmented system maximizes accuracy and explainability, a well-trained domain-adaptive embedding search alone can offer a highly efficient solution for large-scale applications.

REFERENCES

- Auer, S., Bizer, C., Kobilarov, G., Lehmann, J., Cyganiak, R., and Ives, Z. (2007). Dbpedia: A nucleus for a web of open data. In *The Semantic Web (ISWC)*.
- Bengio, Y., Ducharme, R., Vincent, P., and Jauvin, C. (2003). A neural probabilistic language model. *Journal of Machine Learning Research*, 3:1137–1155.
- Bollacker, K., Evans, C., Paritosh, P., Sturge, T., and Taylor, J. (2008). Freebase: A collaboratively created graph database for structuring human knowledge. In *Proceedings of the ACM SIGMOD*.
- Bordes, A., Usunier, N., Garcia-Duran, A., Weston, J., and Yakhnenko, O. (2013). Translating embeddings for modeling multi-relational data. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems (NeurIPS)*, 33.
- Burke, R. (2002). Hybrid recommender systems: Survey and experiments. *User Modeling and User-Adapted Interaction*, 12(4):331–370.
- Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *ICML*.
- Coursera, Inc. (2024). Coursera reports fourth quarter and full year 2023 financial results. <https://investor.coursera.com/news/news-details/2024/Coursera-Reports-Fourth-Quarter-and-Full-Year-2023-Financial-Results/default.aspx>. Accessed: March 02, 2026.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). Bert: Pre-training of deep bidirectional transformers for language understanding. In *NAACL*.
- Dong, H., Long, X., and Li, Y. (2024). Synthetic hard negative samples for contrastive learning. *Neural Processing Letters*, 56:33.
- Enevoldsen, K., Chung, I., Kerboua, I., Kardos, M., Mathur, A., Stap, D., Gala, J., Sibli, W., Krzemiński, D., Winata, G. I., Sturua, S., Utpala, S., Ciancone, M., Schaeffer, M., Sequeira, G., Misra, D., Dhakal, S., Rysstrøm, J., Solomatin, R., Ömer Çağatan, Kundu, A., Bernstorff, M., Xiao, S., Sukhlecha, A., Pahwa, B., Poświata, R., GV, K. K., Ashraf, S., Auras, D., Plüster, B., Harries, J. P., Magne, L., Mohr, I., Hendriksen, M., Zhu, D.,

- Gisserot-Boukhlef, H., Aarsen, T., Kostkan, J., Wojtasik, K., Lee, T., Šuppa, M., Zhang, C., Rocca, R., Hamdy, M., Michail, A., Yang, J., Faysse, M., Vatolin, A., Thakur, N., Dey, M., Vasani, D., Chitale, P., Tedeschi, S., Tai, N., Snegirev, A., Günther, M., Xia, M., Shi, W., Lù, X. H., Clive, J., Krishnakumar, G., Maksimova, A., Wehrli, S., Tikhonova, M., Panchal, H., Abramov, A., Ostendorff, M., Liu, Z., Clematide, S., Miranda, L. J., Fenogenova, A., Song, G., Safi, R. B., Li, W.-D., Borghini, A., Cassano, F., Su, H., Lin, J., Yen, H., Hansen, L., Hooker, S., Xiao, C., Adlakha, V., Weller, O., Reddy, S., and Muennighoff, N. (2025). Mmteb: Massive multilingual text embedding benchmark. *arXiv preprint arXiv:2502.13595*.
- European Commission (2024). European skills, competences, qualifications and occupations (esco). <https://esco.ec.europa.eu/>. Accessed: January 05, 2026.
- Fonseca, J. and Bacao, F. (2023). Tabular and latent space synthetic data generation: a literature review. *Journal of Big Data*, 10(1):115.
- Frej, J. et al. (2024a). Course recommender systems need to consider the job market. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- Frej, J., Shah, N., Knezevic, M., Nazaretsky, T., and Käser, T. (2024b). Finding paths for explainable mooc recommendation: A learner perspective. In *Proceedings of the 14th Learning Analytics and Knowledge Conference*, page 426–437. ACM.
- Gao, L., Zhang, Y., Han, J., and Callan, J. (2021). Scaling deep contrastive learning batch size under memory limited setup. In *Proceedings of the 6th Workshop on Representation Learning for NLP (RepL4NLP-2021)*, pages 316–321, Online. Association for Computational Linguistics.
- Gheewala, S., Xu, S., and Yeom, S. (2025). In-depth survey: deep learning in recommender systems—exploring prediction and ranking models. *Neural Computing and Applications*.
- Hogan, A., Blomqvist, E., Cochez, M., d’Amato, C., Melo, G. d., Gutierrez, C., et al. (2021). Knowledge graphs. *ACM Computing Surveys*, 54(4):1–37.
- Hu, H., Pan, L., Ran, Y., and Kan, M.-Y. (2022). Modeling and leveraging prerequisite context in recommendation. In *Proceedings of Workshop on Context-Aware Recommender Systems (CARS @ RecSys 2022)*.
- Huang, Y., Wang, K., Dutta, S., Patel, R. N., Glavaš, G., and Gurevych, I. (2023). Adasent: Efficient domain-adapted sentence embeddings for few-shot classification.

In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Ji, S., Pan, S., Cambria, E., Marttinen, P., and Yu, P. S. (2021). A survey on knowledge graphs: Representation, acquisition, and applications. *IEEE Transactions on Neural Networks and Learning Systems*.

Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., and Yih, W.-t. (2020). Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781. Association for Computational Linguistics.

Koren, Y., Bell, R., and Volinsky, C. (2009). Matrix factorization techniques for recommender systems. *Computer*, 42(8):30–37.

LinkedIn Learning (2025). 2025 workplace learning report: The next era of learning and development. Research report, LinkedIn Corporation. <https://learning.linkedin.com/resources/workplace-learning-report> (Accessed: March 2, 2026).

LM-Kit (2025). Qwen3-14b-instruct - GGUF. <https://huggingface.co/lm-kit/qwen-3-14b-instruct-gguf>. Accessed: February 05, 2026.

Lops, P., de Gemmis, M., and Semeraro, G. (2011). Content-based recommender systems: State of the art and trends. *Recommender Systems Handbook*, pages 73–105.

Mikolov, T., Chen, K., Corrado, G., and Dean, J. (2013). Efficient estimation of word representations in vector space. In *Proceedings of the 1st International Conference on Learning Representations*.

Moreira, G. d. S. P., Osmulski, R., Xu, M., Ak, R., Schifferer, B., and Oldridge, E. (2024). Nv-retriever: Improving text embedding models with effective hard-negative mining. *arXiv preprint arXiv:2407.15831*.

Naveed, H., Khan, A. U., Qiu, S., Saqib, M., Anwar, S., Usman, M., Akhtar, N., Barnes, N., and Mian, A. (2025). A comprehensive overview of large language models. *ACM Transactions on Intelligent Systems and Technology*, 16(5):1–72.

Nesterowicz, K., Bayramova, U., Fereshtehnejad, S.-M., Scarlat, A., Ash, A., Augustyn, A. M., and Szádeczky, T. (2022). Gamification increases completion rates in massive open online courses. *International Journal of Information and Communication Technology Education (IJICTE)*, 18(1):1–15.

- Nickel, M., Murphy, K., Tresp, V., and Gabrilovich, E. (2015). A review of relational machine learning for knowledge graphs. *Proceedings of the IEEE*, 104(1):11–33.
- Pennington, J., Socher, R., and Manning, C. D. (2014). Glove: Global vectors for word representation. In *EMNLP*.
- Peters, M. E., Neumann, M., Iyyer, M., Gardner, M., Clark, C., Lee, K., and Zettlemoyer, L. (2018). Deep contextualized word representations. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 2227–2237. Association for Computational Linguistics. Accessed: January 28, 2026.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. (2021). Learning transferable visual models from natural language supervision. *CoRR*, abs/2103.00020.
- Reimers, N. and Gurevych, I. (2019). Sentence-bert: Sentence embeddings using siamese bert-networks. In *EMNLP*.
- Sabet, A. J., Bana, S. H., Yu, R., and Frank, M. R. (2024). Course-skill atlas: A national longitudinal dataset of skills taught in u.s. higher education curricula. *Scientific Data*, 11(1):1086.
- Shah, D. (2021). By the numbers: Moocs in 2021. <https://www.classcentral.com/report/mooc-stats-2021/>. Accessed: January 10, 2026.
- Sun, J., Mei, S., Yuan, K., Jiang, Y., and Cao, J. (2024). Prerequisite-enhanced category-aware graph neural networks for course recommendation. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 18(5):1–21.
- Sun, Z., Deng, Z.-H., Nie, J.-Y., and Tang, J. (2019). Rotate: Knowledge graph embedding by relational rotation in complex space. In *International Conference on Learning Representations (ICLR)*.
- Turchin, A. et al. (2023). Comparison of bert implementations for medical text. *Informatics in Medicine Unlocked*, 39:101235.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30.
- Wang, Q., Mao, Z., Wang, B., and Guo, L. (2017). Knowledge graph embedding: A survey of approaches and applications. *IEEE Transactions on Knowledge and Data Engineering*, 29(12):2724–2743.

- Wang, X., He, X., Wang, D., Feng, F., and Chua, T.-S. (2019a). Kgat: Knowledge graph attention network for recommendation. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 950–958.
- Wang, X., Wang, D., Xu, C., He, X., Cao, Y., and Chua, T.-S. (2019b). Explainable reasoning over knowledge graphs for recommendation. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- World Economic Forum (2023). The future of jobs report 2023. Technical report, World Economic Forum, Geneva, Switzerland.
- Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., Zheng, C., Liu, D., Zhou, F., Huang, F., Hu, F., Ge, H., Wei, H., Lin, H., Tang, J., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Zhou, J., Lin, J., Dang, K., Bao, K., Yang, K., Yu, L., Deng, L., Li, M., Xue, M., Li, M., Zhang, P., Wang, P., Zhu, Q., Men, R., Gao, R., Liu, S., Luo, S., Li, T., Tang, T., Yin, W., Ren, X., Wang, X., Zhang, X., Ren, X., Fan, Y., Su, Y., Zhang, Y., Zhang, Y., Wan, Y., Liu, Y., Wang, Z., Cui, Z., Zhang, Z., Zhou, Z., and Qiu, Z. (2025). Qwen3 technical report.
- Zhang, F., Yuan, N. J., Lian, D., Xie, X., and Huang, S. (2016). Collaborative knowledge base embedding for recommender systems. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Francisco, CA, USA, August 13-17, 2016*, pages 353–362.
- Zhang, Y., Li, M., Long, D., Zhang, X., Lin, H., Yang, B., Xie, P., Yang, A., Liu, D., Lin, J., Huang, F., and Zhou, J. (2025). Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*.

APPENDIX

A Prompts

A.1 Training dataset Generation Prompt

The training dataset generation prompt (Figure A.1) is employed to create diverse linguistic variations of a single ESCO skill. Simultaneously, it instructs the model to produce "hard negatives"—skills that are contextually related but conceptually distinct. This dual-task approach is critical for training the embedding model to distinguish between subtle professional nuances in academic discourse.

```

You are an expert in curriculum design and skill taxonomy.

Given this skill from ESCO:
- Skill Name: {skill_name}
- Description: {full_description}

Task 1: Generate {n} Variations
Generate diverse ways this skill appears in university syllabi. Each
variation should:
- Use natural language and synonyms.
- Vary in length (short phrases to full sentences).
- IMPORTANT: Must represent the EXACT SAME skill.

Task 2: Generate {m} Hard Negatives
Generate skill descriptions that are similar in wording but describe
DIFFERENT skills:
- Related but not the same (e.g., "Software debugging" vs "Software
testing").
- Confusing or subtly different (wrong domain or application).
- IMPORTANT: These must be challenging enough to confuse an AI model.

Output Format:
Return ONLY a JSON object with this structure:
{
  "variations": ["variation 1", "variation 2", ...],
  "hard_negatives": ["negative 1", "negative 2", ...]
}

```

Figure A.1 Prompt for Generating Semantic Variations and Challenging Hard Negatives

A.2 Candidate Profile Generation Prompt

The candidate profile generation prompt (Figure A.2) is designed to simulate a recruitment scenario where a synthetic candidate is created with a specific, controlled level of mismatch (60%–80%) against a given job description. This ensures that the evaluation dataset contains realistic "skills gaps" that the recommendation system must subsequently address.

You are an expert HR professional. Your task is to generate a candidate profile that matches approximately {matching_percent}% with the following job description.

Job Description:

- Position: {jd_info['title']}
- Description: {jd_info['description']}
- Required Technical Skills: {jd_info['technical_skills']}
- Required Soft Skills: {jd_info['soft_skills']}
- Experience Required: {jd_info['experience']}
- Degree Required: {jd_info['degree']}
- Keywords: {jd_info['keywords']}

Matching Guidelines:

1. If matching is 60-65%: Candidate has ~60% technical/soft skills; experience and degree level may be lower than required.
2. If matching is 66-74%: Candidate has ~70% technical/soft skills; experience and degree are close to requirements.
3. If matching is 75-80%: Candidate has 75-80% technical skills; experience and degree match requirements; most soft skills present.

IMPORTANT Instructions:

- Generate realistic candidate data with deliberate gaps.
- DO NOT make a 100% match.
- Return arrays for technical_skills and soft_skills (skills the candidate HAS).
- Return strings for experience (e.g., "2 years") and degree (e.g., "Bachelor").
- Identify skill_gaps clearly:
 - * missing_technical_skills: Required by JD but NOT in candidate profile.
 - * missing_soft_skills: Required by JD but NOT in candidate profile.
 - * experience_gap: Describe missing years or education shortfall.

Figure A.2 Prompt for Generating Synthetic Candidate Profiles with Controlled Mismatch

A.3 Course Extraction Prompt

The course extraction prompt (Figure A.3) is designed to guide the Large Language Model in processing unstructured Vietnamese course syllabi. It focuses on decomposing Course Learning Outcomes (CLO) and course content into standardized, professional skill sets categorized by type (Hard Skill, Soft Skill, or Tool).

You are an expert Job Analyst and HR Specialist.

Your task is to analyze Vietnamese course syllabus documents and extract structured data.

=== SKILL EXTRACTION RULES ===

1. Extract ALL distinct skills - do NOT merge separate skills into one.

BAD: 'Cryptography' covering both symmetric AND asymmetric encryption.

GOOD: Two separate entries: 'Symmetric Encryption' and 'Public Key Cryptography'.

2. GENERALIZE academic phrasing into standard professional skill names (1-4 words).

BAD: 'Detect, analyze and evaluate weaknesses of network systems'.

GOOD: 'Network Vulnerability Assessment'.

3. PRIORITIZE the CLO table (Chuẩn đầu ra học phần / Course Learning Outcomes) as the primary source of skills. Each CLO row typically maps to one or more skills.

4. Also extract skills from: course description (Mô tả học phần), main content (Nội dung chính), and weekly lesson plan (Kế hoạch dạy học).

=== CATEGORY CLASSIFICATION ===

- Hard Skill: domain knowledge, technical or academic competencies.

Examples: Network Security, Cryptography, SQL, Machine Learning.

- Soft Skill: interpersonal, cognitive, or behavioral competencies.

Examples: Teamwork, Critical Thinking, Technical Writing, Self-learning.

- Tool/Software: a named tool, platform, language, or software product.

Examples: Wireshark, Python, Solidity, Ethereum, MySQL.

Output in ENGLISH.

Figure A.3 Skill Extraction Prompt for Processing Vietnamese Course Syllabi

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